CHAPTER 08

Carboxylic Acids

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: 1a683d2b84234d2d686caa9dc0651b55fd211ada11f402c068a812d3b5989dc5

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

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ch08:lesson:soap | soap:step-1, soap:step-2

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ch08:lesson:soap:cue | soap:cue

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ch08:lesson:soap:app | soap:app

ch08:molecule:aceticacid | aceticacid

ch08:molecule:acetate | acetate

ch08:molecule:ester | ester

ch08:molecule:acetamide | acetamide

ch08:molecule:acetoaceticacid | acetoaceticacid

ch08:molecule:acetone | acetone

ch08:molecule:laurate | laurate

ch08:curriculum | scope-1, scope-2

Curriculum scope

scope-1: 8.1 Description and Properties

scope-2: 8.2 Reactions of Carboxylic Acids

Objective: Predict acid ionization and physical properties

Objective: Explain esterification, activation, and reduction

Objective: Track decarboxylation and saponification

Supplied Kaplan curriculum: Chapter 8; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter08b.xhtml

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

acid-properties | Carboxylic acid structure and acidity

Two oxygens stabilize the conjugate base

acid-properties:rule | Core Rules

A carboxyl group has C=O and OH bonded to the same carbon. Carboxylic acids hydrogen bond and can dimerize, especially in nonpolar media. Their carboxylates are resonance stabilized. Electron-withdrawing substituents generally increase acidity, with inductive effects weakening over distance.

Visual card: Carboxylic Acid Centrosymmetric Dimers, Resonance Acidity & Inductive Distance Decay

Chemistry diagram: Neutral acid versus carboxylate. The complete graphs retain all formal charges.

$RCOOH + NaHCO_3 \rightarrow RCOO^- Na^+ + H_2O + CO_2(g)\uparrow$ (Diagnostic Fizzing Test for pKa < 6)

Carboxylic Acid pKa ~4–5

8-Membered H-Bond Dimer → ↑↑ BP

EWG (CF_3, Cl, NO_2) → ↓ pKa

Dicarboxylic: pKa1 < pKa2

Alcohol (ROH, pKa ~16) | Negative charge localized on 1 O | Insoluble in $NaHCO_3$ and NaOH

Phenol (ArOH, pKa ~10) | Delocalized over 1 O + 3 ring C atoms | Soluble in NaOH, insoluble in $NaHCO_3$

Carboxylic Acid (pKa ~4.8) | Delocalized equally over TWO O atoms | Soluble in weak $NaHCO_3$ (evolves $CO_2\uparrow$ gas!)

Dimeric Hydrogen Bonding — — Pairs of carboxylic acids form cyclic 8-membered rings with TWO O–H···O=C hydrogen bonds, giving higher boiling points than alcohols of the same MW.

Inductive Substituent Effects — — Electron-withdrawing groups stabilize $RCOO^-$ and lower pKa: CF_3COOH (pKa 0.2) < CCl_3COOH (0.7) < $ClCH_2COOH$ (2.9) < CH_3COOH (4.76). Effect decays rapidly with distance (α > β > γ)!

acid-properties:cue | Memory Cues

pH above pKa → more carboxylate. pH below pKa → more neutral acid.

Visual card: Ranking Carboxylic Acid Acidity (pKa) on the MCAT

What differs between the carboxylic acids you are ranking?

Halogen Electronegativity & Count — More Halogens & Higher EN → Lower pKa (Stronger Acid) — FCH_2COOH (pKa 2.6) is stronger than $ClCH_2COOH$ (2.9) > $BrCH_2COOH$ (2.9); Cl_3CCOOH (0.7) is much stronger than $ClCH_2COOH$ (2.9).

Distance from –COOH (α vs. β vs. γ) — Closer to –COOH → Stronger Inductive Pull (α > β > γ) — 2-chlorobutanoic acid (pKa 2.8) > 3-chlorobutanoic acid (4.0) > 4-chlorobutanoic acid (4.5) > butanoic acid (4.8).

Dicarboxylic Acids (HOOC–(CH_2)_n–COOH) — pKa1 Is Lower Than 4.8; pKa2 Is Higher Than 4.8! — The neutral second –COOH acts as an inductive EWG to lower pKa1; once the first $–COO^-$ forms, its −1 charge repels loss of the second H^+ (raising pKa2).

acid-properties:trap | Common Mistakes / MCAT Traps

The OH of COOH is not an ordinary alcohol group. When a carbon ring bears a COOH substituent, the carboxyl carbon is outside the ring, attached to a ring carbon. A lactone can have an endocyclic acyl carbon, but its group is a cyclic ester, not free COOH. Larger hydrocarbon regions can still limit aqueous solubility.

Visual card: Drawing Carboxylate Resonance by Breaking the C–C Single Bond or Using Periodic Column Size for Inductive Effect

Common Trap — — Saying I–CH_2COOH is stronger than F–CH_2COOH because iodine is larger.

Correct Reasoning — — In inductive effects (H is NOT attached to the halogen!), ONLY electronegativity matters: F > Cl > Br > I (size only dominates when the acidic H is bonded directly to X, as in HI vs HF)!

acid-properties:app | MCAT Applications

Ionization controls how fatty acids, drug acids, and amino-acid side chains partition between environments.

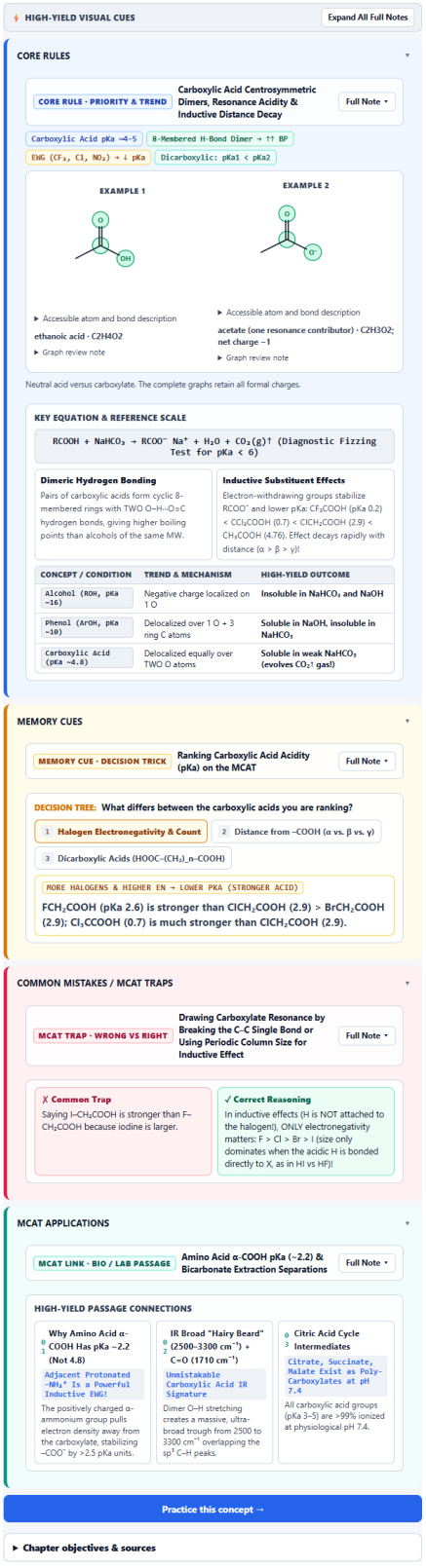
Visual card: Amino Acid α-COOH pKa (~2.2) & Bicarbonate Extraction Separations

Why Amino Acid α-COOH Has pKa ~2.2 (Not 4.8) — Adjacent Protonated $–NH_3^+$ Is a Powerful Inductive EWG! — The positively charged α-ammonium group pulls electron density away from the carboxylate, stabilizing $–COO^-$ by >2.5 pKa units.

IR Broad "Hairy Beard" (2500–3300 cm^{-1}) + C=O (1710 cm^{-1}) — Unmistakable Carboxylic Acid IR Signature — Dimer O–H stretching creates a massive, ultra-broad trough from 2500 to 3300 cm^{-1} overlapping the sp^3 C–H peaks.

Citric Acid Cycle Intermediates — Citrate, Succinate, Malate Exist as Poly-Carboxylates at pH 7.4 — All carboxylic acid groups (pKa 3–5) are >99% ionized at physiological pH 7.4.

Screenshot: acid-properties-visuals



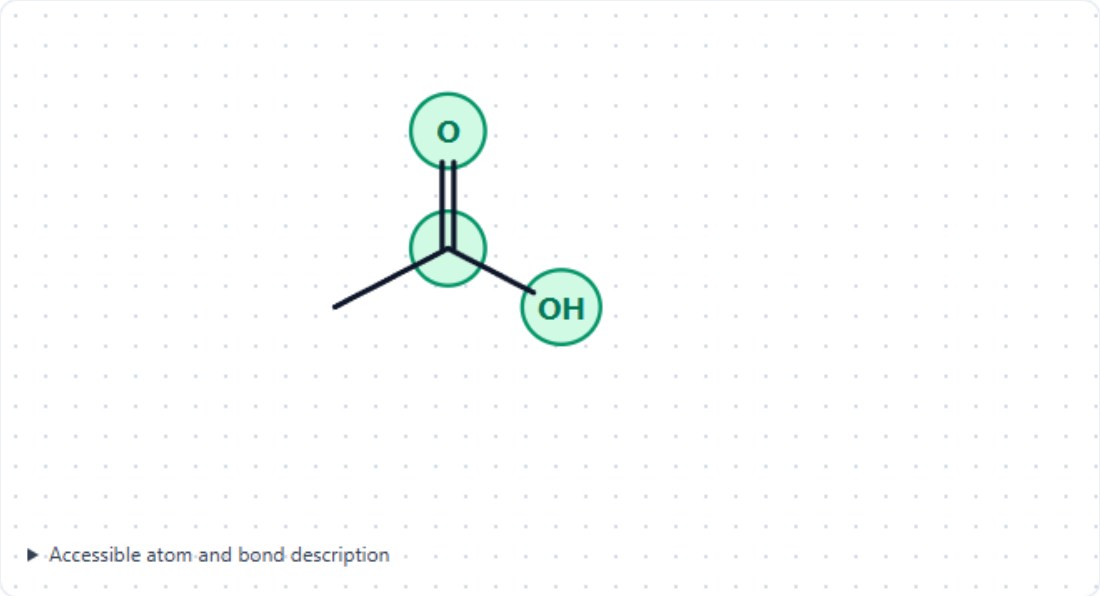
acid-properties:step-1 | 1. Recognize COOH

The acid functional group includes both oxygens and their shared carbon. Its carbon has no room for another carbon substituent beyond the existing skeleton.

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Screenshot: acid-properties-step-1



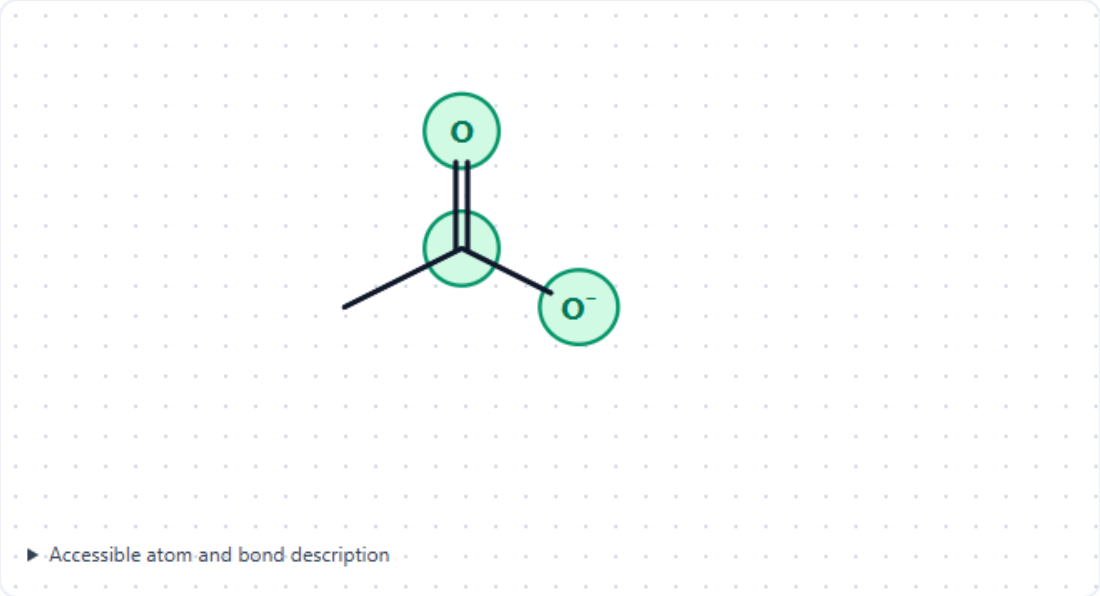
acid-properties:step-2 | 2. Change pH

Adjust pH around a model pKa of 4.8. At pH=pKa, [A−]=[HA]; two units above pKa gives a 100:1 ratio, about 99% A−.

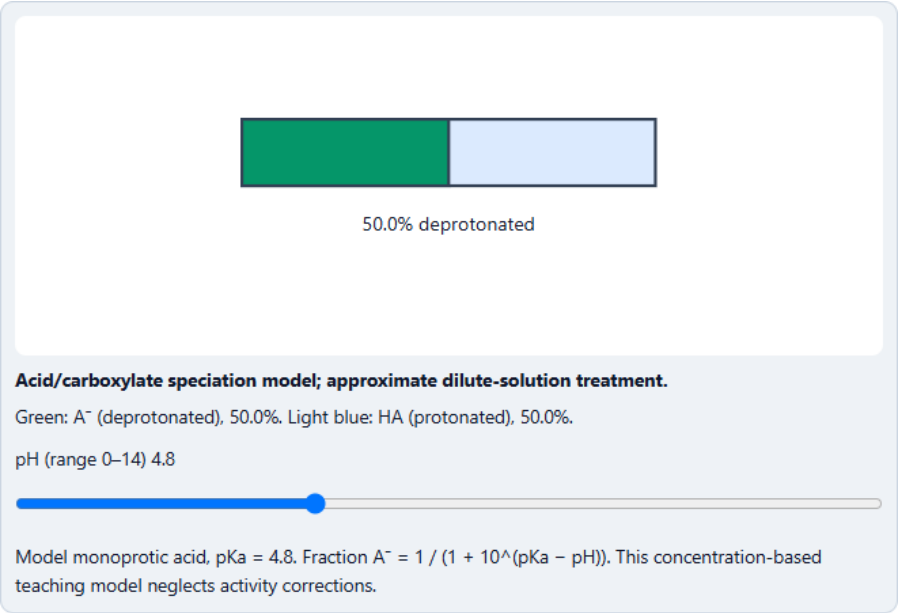
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Screenshot: acid-properties-step-2



Screenshot: acid-properties-step-2-diagram



{"diagramSpecification": {"type": "acid", "pKa": 4.8, "caption": "Acid/carboxylate speciation model; approximate dilute-solution treatment."}}

acid-reactions | Formation and acyl substitution

Activate the acid or use appropriate catalysis

acid-reactions:rule | Core Rules

Primary alcohols and aldehydes can be oxidized to acids. Fischer esterification combines acid and alcohol under acid catalysis; removing water or using excess alcohol drives the reversible reaction. Direct acid plus amine usually first gives an ammonium carboxylate salt; efficient amide synthesis generally requires activation or dehydrating conditions.

Visual card: Carboxylic Acid Activation ($SOCl_2$, DCC/EDC), Fischer Esterification & Acyl Substitution

Chemistry diagram: Separate acid and ester examples; different carbon skeletons, not an esterification pair.

$SOCl_2$ → Acid Chloride (RCOCl)

RCOOH + ROH + H^+ ⇌ Ester (Fischer)

RCOOH + R′NH_2 + Δ → Amide (Need EDC/DCC at RT!)

Acid + Amine (Room Temp) | Acid–base reaction ONLY! | Forms unreactive carboxylate ammonium salt ($RCOO^-$ R′NH_3^+)

DCC / EDC Coupling Agent | Activates –OH into good leaving group | Forms amide bond at room temperature without heat

Fischer Esterification | RCOOH + R′OH + Cat. H_2SO_4 ⇌ Ester + H_2O | Acyl C–O cleaves; alcohol O* ends up in ester!

Addition–Elimination (Nucleophilic Acyl Substitution) — — Unlike aldehydes/ketones (which lack a leaving group), carboxylic acids and derivatives undergo nucleophilic addition to a tetrahedral intermediate followed by ELIMINATION of the leaving group to restore C=O.

Converting –OH Into a Top-Tier Chloride ($SOCl_2$ or PCl_5) — — Unactivated –OH is a poor leaving group; thionyl chloride ($SOCl_2$) converts RCOOH into highly reactive acid chloride RCOCl (driven by irreversible SO_2↑ + HCl↑ gas evolution).

Direct Amide Coupling Trap (Acid + Amine) — — Mixing RCOOH + R′NH_2 immediately transfers a proton to give an unreactive salt ($RCOO^-$ R′NH_3^+); forming an amide at room temp requires activating –COOH first with $SOCl_2$ or carbodiimides (DCC/EDC)!

acid-reactions:cue | Memory Cues

Proton transfer is often faster than acyl substitution. Check it before drawing a new amide bond.

Visual card: Isotope Tracking (^{18}O) & Equilibrium Control in Fischer Esterification

What question is asked about Fischer esterification (RCOOH + R′^{18}OH ⇌ RCO^{18}OR′ + H_2O)?

Where Does the ^{18}O Label on the Alcohol Go? — 100% into the Ester (R–CO–^{18}OR′); Water Gets Unlabeled H_2O! — The alcohol oxygen attacks the carbonyl carbon; the carboxylic acid –OH is protonated twice to leave as unlabeled H_2O.

How Do You Drive Reversible Fischer Esterification (K_{eq} ~ 1–4) to 100%? — Add Excess Alcohol OR Remove H_2O (Dean-Stark / Molecular Sieves) — By Le Chatelier's principle, removing water or using the alcohol as solvent shifts equilibrium toward the ester.

Why Does tert-Butanol Fail in Standard Acid Fischer Esterification? — t-BuOH Forms a 3° Carbocation in Acid → Eliminates to Isobutylene! — Make tert-butyl esters by reacting t-BuOH with an acid chloride + pyridine (or reacting RCOOH with isobutylene).

acid-reactions:trap | Common Mistakes / MCAT Traps

OH− is a poor leaving group. Acid catalysis can enable water loss; coupling reagents or reactive acid derivatives can enable amide formation. Do not draw spontaneous amide synthesis just by mixing acid and amine in water.

Visual card: Assuming RCOOH + RNH_2 Forms an Amide Directly at Room Temperature

Common Trap — — Drawing a peptide/amide bond directly from RCOOH + R′NH_2 in water at 25 °C without a coupling reagent.

Correct Reasoning — — Amines (pKaH ~10) simply deprotonate carboxylic acids (pKa ~5) to form a stable carboxylate-ammonium salt; use EDC/DCC, $SOCl_2$, or biological ATP/aminoacyl-tRNA activation!

acid-reactions:app | MCAT Applications

Peptide synthesis uses activated carboxyl groups; enzyme-catalyzed biosynthesis likewise supplies activation and organization.

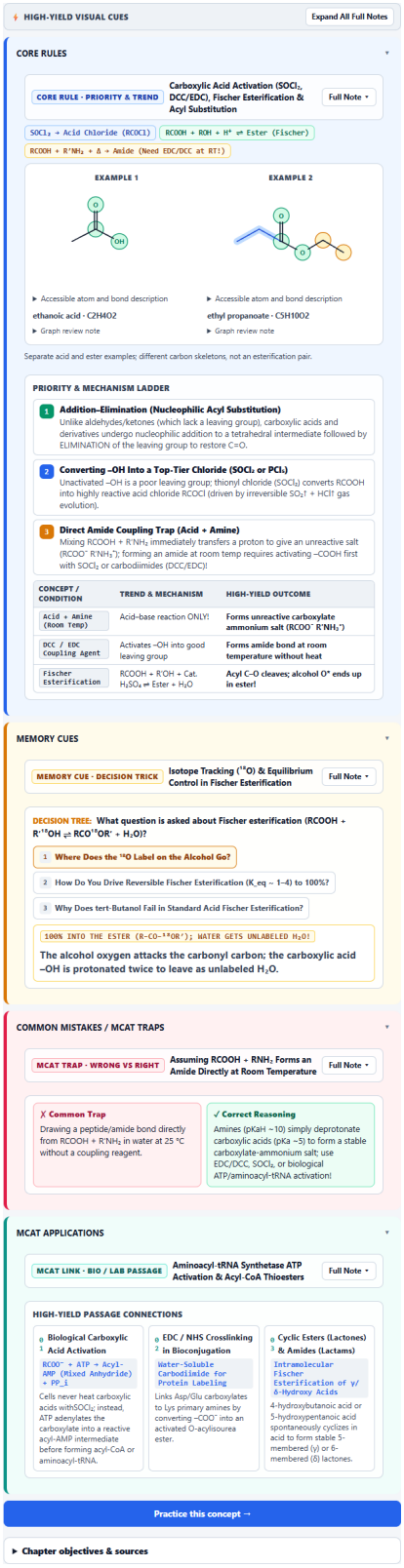
Visual card: Aminoacyl-tRNA Synthetase ATP Activation & Acyl-CoA Thioesters

Biological Carboxylic Acid Activation — $RCOO^-$ + ATP → Acyl-AMP (Mixed Anhydride) + PP_i — Cells never heat carboxylic acids with$SOCl_2$; instead, ATP adenylates the carboxylate into a reactive acyl-AMP intermediate before forming acyl-CoA or aminoacyl-tRNA.

EDC / NHS Crosslinking in Bioconjugation — Water-Soluble Carbodiimide for Protein Labeling — Links Asp/Glu carboxylates to Lys primary amines by converting $–COO^-$ into an activated O-acylisourea ester.

Cyclic Esters (Lactones) & Amides (Lactams) — Intramolecular Fischer Esterification of γ/δ-Hydroxy Acids — 4-hydroxybutanoic acid or 5-hydroxypentanoic acid spontaneously cyclizes in acid to form stable 5-membered (γ) or 6-membered (δ) lactones.

Screenshot: acid-reactions-visuals



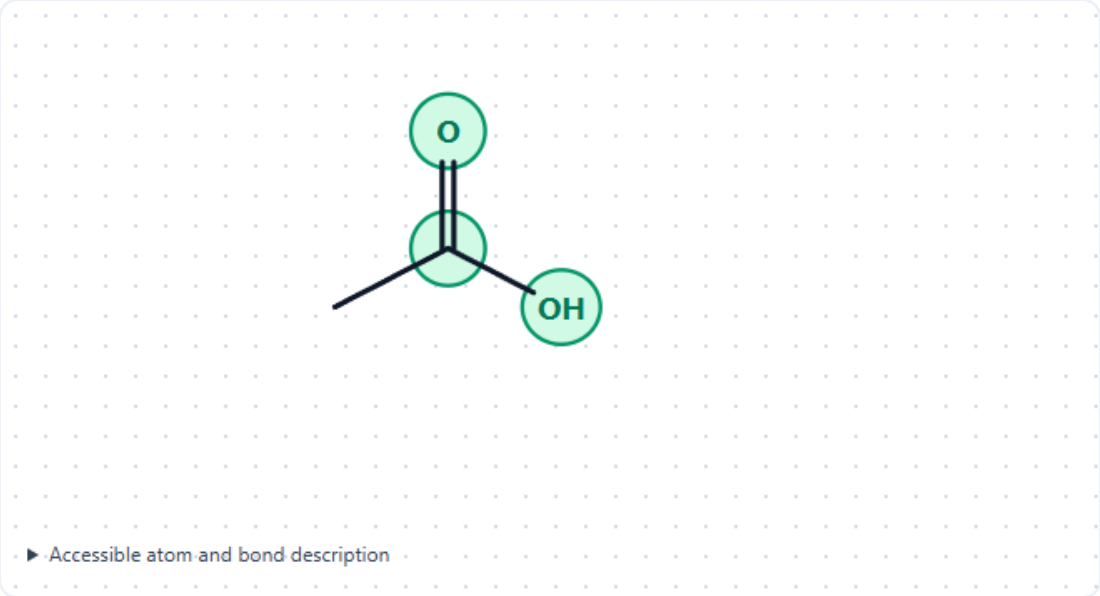
acid-reactions:step-1 | 1. Acid substrate

The electrophilic acyl carbon is the carbonyl carbon. Protonating carbonyl oxygen increases electrophilicity in acid-catalyzed esterification.

Displayed molecule ID: aceticacid

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Screenshot: acid-reactions-step-1



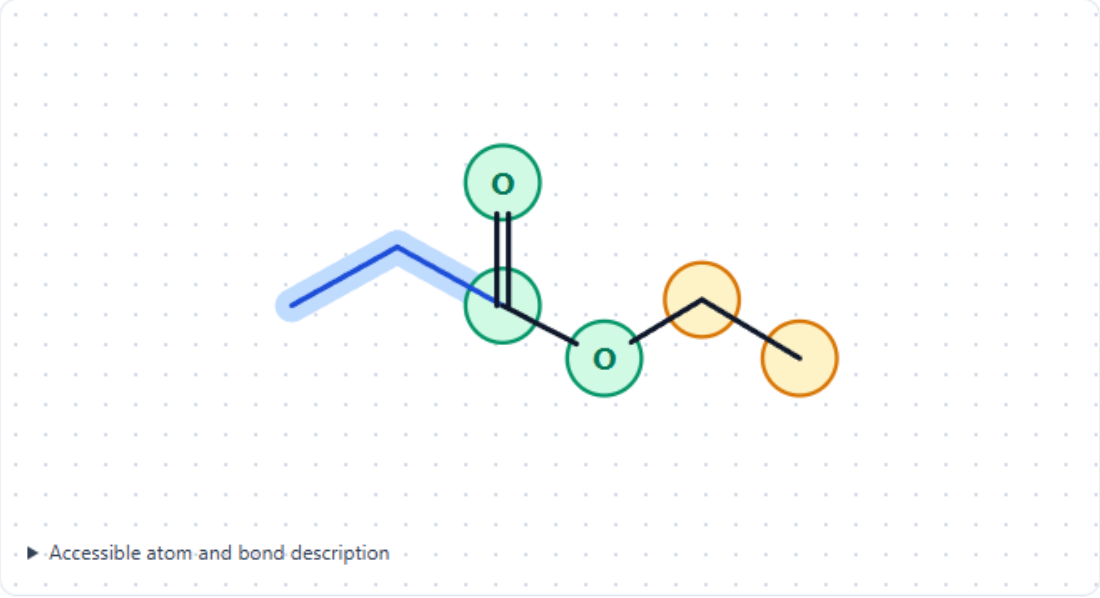
acid-reactions:step-2 | 2. Ester product pattern

An ester has an OR group where the acid had OH. This example is ethyl propanoate: the O-ethyl and three-carbon acyl portions have different origins.

Displayed molecule ID: ester

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Screenshot: acid-reactions-step-2



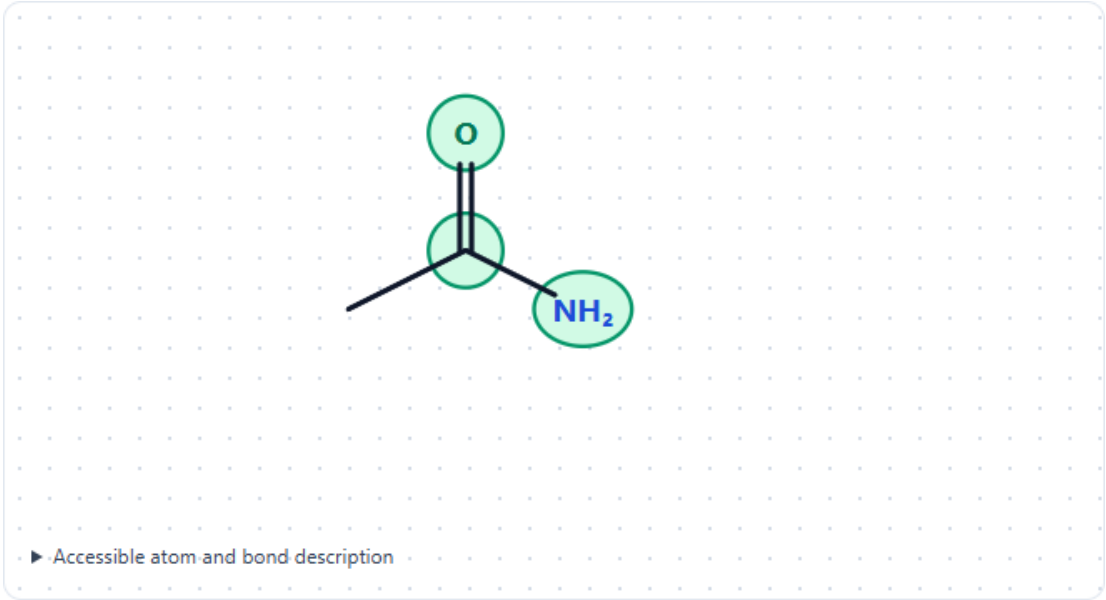
acid-reactions:step-3 | 3. Amide product pattern

An amide has N attached directly to the acyl carbon. Formation requires conditions that overcome acid–base salt formation and poor leaving-group ability.

Displayed molecule ID: acetamide

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Screenshot: acid-reactions-step-3



reduction-decarboxylation | Reduction and beta-keto-acid decarboxylation

Track carbon atoms, not just functional groups

reduction-decarboxylation:rule | Core Rules

LiAlH4 followed by appropriate aqueous workup reduces a carboxylic acid to a primary alcohol. Ordinary NaBH4 conditions generally do not. Heating a beta-keto acid can release CO2 through a favorable cyclic transition state, initially giving an enol that tautomerizes to a carbonyl.

Visual card: Carboxylic Acid Reduction ($LiAlH_4$ / BH_3) & Thermal β-Keto Acid Decarboxylation

Chemistry diagram: Heating a suitable beta-keto acid can release CO2, followed by tautomerization; schematic net pathway.

RCOOH + $LiAlH_4$ (or BH_3) → 1° Alcohol

$NaBH_4$ = NO Reaction with RCOOH

β-Keto Acid + Δ → Ketone + CO_2↑

Simple RCOOH + Heat | Stable up to >200 °C | Does NOT decarboxylate (would form unstable R^- carbanion)

β-Keto Acid / Malonic Acid + Δ | Loses CO_2↑ at mild heat (50–100 °C) | Cyclic 6-membered TS → Enol ⇌ Methyl Ketone

RCOOH + $LiAlH_4$ (then H_3O^+) | Reduces all the way to 1° Alcohol (RCH_2OH) | Aldehyde intermediate is more reactive than acid (cannot stop!)

Complete Reduction to 1° Alcohol ($LiAlH_4$ or BH_3·THF) — — $LiAlH_4$ reduces RCOOH all the way to RCH_2OH; BH_3 (borane) selectively reduces carboxylic acids even in the presence of ketones or nitro groups!

Requirement for Thermal Decarboxylation (C=O at the β-Position!) — — A carboxylic acid loses CO_2 upon gentle heating ONLY when a carbonyl (ketone, aldehyde, or second –COOH) sits at the β-carbon (3-oxocarboxylic acid / 1,3-dicarbonyl).

6-Membered Cyclic Transition State → Enol → Ketone — — Intramolecular H-bond between the –COOH proton and the β-C=O oxygen enables a concerted 6-electron pericyclic ring shift, releasing CO_2 gas and an enol that tautomerizes to a ketone.

reduction-decarboxylation:cue | Memory Cues

Reduction retains the acid carbon. Decarboxylation loses it as CO2.

Visual card: Will This Carboxylic Acid Decarboxylate Upon Heating?

Count from the carboxylic acid carbon (C1) to the other carbonyl:

Carbonyl at C3 (β-Keto Acid or Malonic Acid, 1,3-Relationship) — YES! Loses CO_2↑ Easily Upon Heating — Replace the entire –COOH group with –H! (Example: acetoacetic acid → acetone + CO_2↑; malonic acid → acetic acid + CO_2↑).

Carbonyl at C2 (α-Keto Acid) or C4 (γ-Keto Acid) — NO Thermal Decarboxylation Without an Enzyme — Cannot form the 6-membered cyclic transition state or stabilize the resulting α-enol/enolate; α-keto acids require TPP cofactor in biology.

Bridgehead Bicyclic β-Keto Acid (Bredt's Rule Trap!) — NO Decarboxylation! (Bridgehead Cannot Form Planar Enol C=C) — Decarboxylation requires forming a Cα=Cβ enol double bond; if Cα is a rigid bicyclic bridgehead, Bredt's rule forbids a bridgehead alkene!

reduction-decarboxylation:trap | Common Mistakes / MCAT Traps

A beta-keto acid has a ketone at the beta position relative to the acid carbon. Not every carboxylic acid readily decarboxylates on heating. The six-membered arrangement is a transition state, not a stable ring intermediate.

Visual card: Decarboxylating Any Random Carboxylic Acid or Stopping $LiAlH_4$ at the Aldehyde

Common Trap — — Removing CO_2 from hexanoic acid or 4-oxopentanoic acid just because "Δ" is written over the arrow.

Correct Reasoning — — Only β-keto acids (3-oxo acids) and malonic acids (1,3-diacids) undergo spontaneous thermal decarboxylation via the 6-membered cyclic enol pathway!

reduction-decarboxylation:app | MCAT Applications

Decarboxylation changes carbon count, a useful check when following metabolic or synthetic pathways.

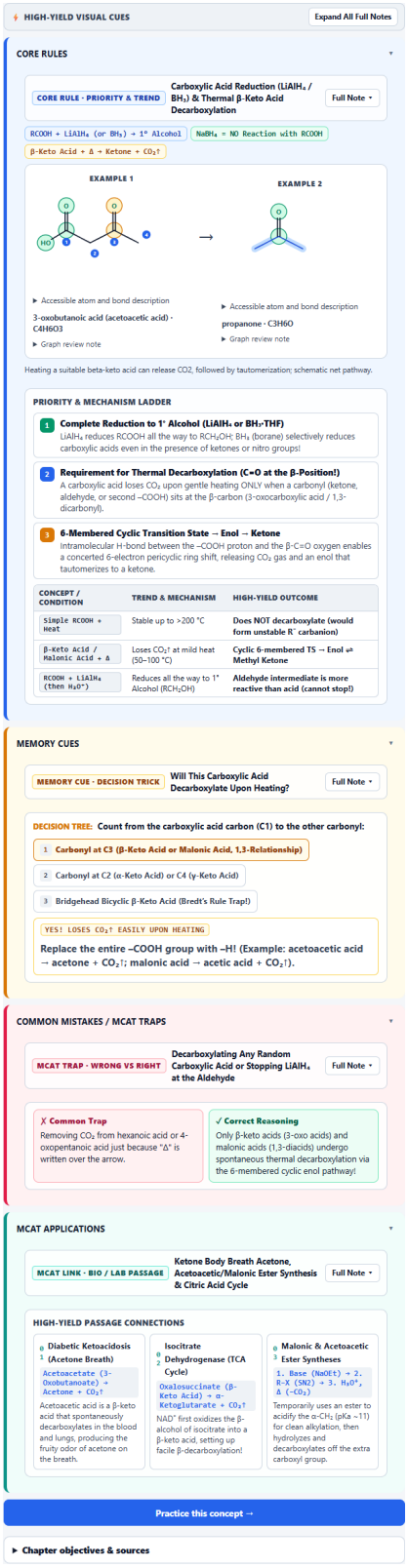
Visual card: Ketone Body Breath Acetone, Acetoacetic/Malonic Ester Synthesis & Citric Acid Cycle

Diabetic Ketoacidosis (Acetone Breath) — Acetoacetate (3-Oxobutanoate) → Acetone + CO_2↑ — Acetoacetic acid is a β-keto acid that spontaneously decarboxylates in the blood and lungs, producing the fruity odor of acetone on the breath.

Isocitrate Dehydrogenase (TCA Cycle) — Oxalosuccinate (β-Keto Acid) → α-Ketoglutarate + CO_2↑ — NAD^+ first oxidizes the β-alcohol of isocitrate into a β-keto acid, setting up facile β-decarboxylation!

Malonic & Acetoacetic Ester Syntheses — 1. Base (NaOEt) → 2. R–X (SN2) → 3. H_3O^+, Δ (−CO_2) — Temporarily uses an ester to acidify the α-CH_2 (pKa ~11) for clean alkylation, then hydrolyzes and decarboxylates off the extra carboxyl group.

Screenshot: reduction-decarboxylation-visuals



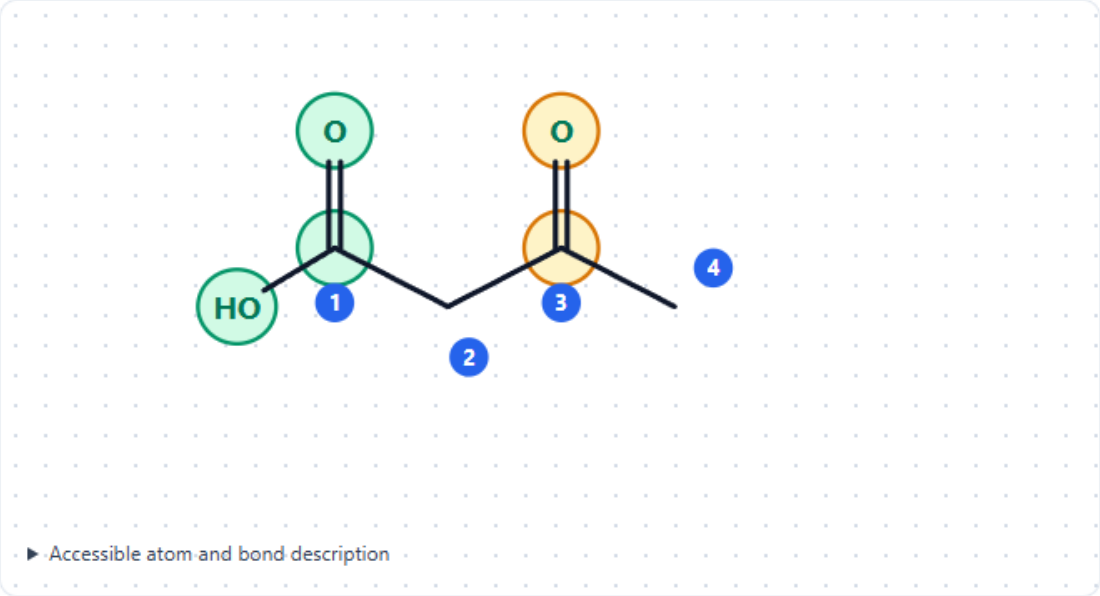
reduction-decarboxylation:step-1 | 1. Number the beta-keto acid

C1 is the carboxyl carbon, C2 is alpha, and C3 is the ketone-bearing beta carbon.

Displayed molecule ID: acetoaceticacid

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Screenshot: reduction-decarboxylation-step-1



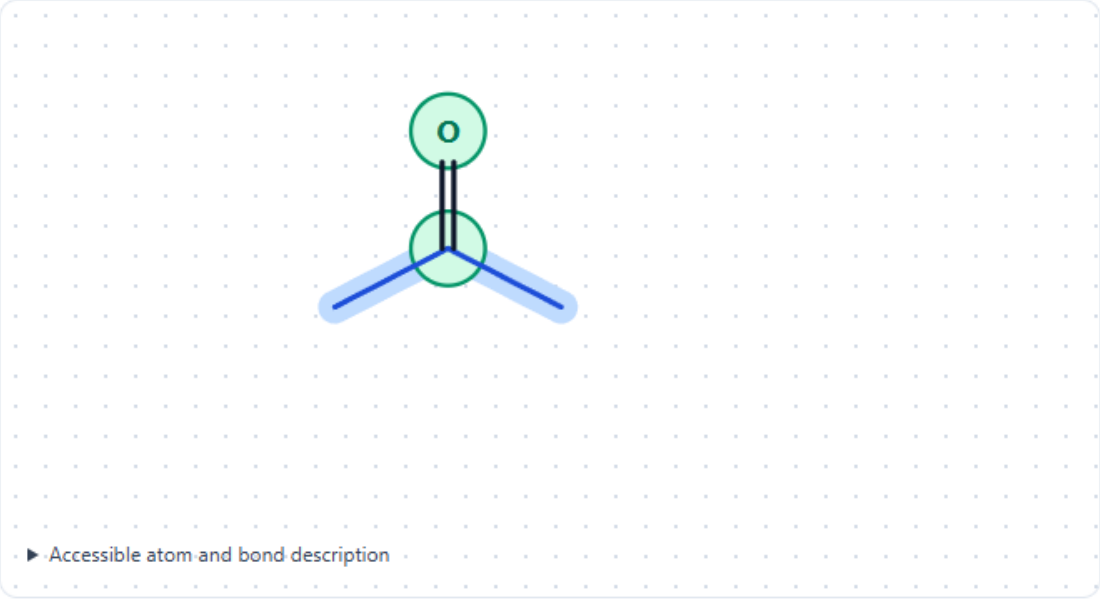
reduction-decarboxylation:step-2 | 2. Lose the acid carbon

The carboxyl carbon leaves as CO2. The remaining three-carbon fragment initially forms an enol, then becomes acetone by tautomerization.

Displayed molecule ID: acetone

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Screenshot: reduction-decarboxylation-step-2



soap | Saponification and amphiphiles

Hydrolysis produces carboxylate salts

soap:rule | Core Rules

Saponification is base-promoted ester hydrolysis, including hydrolysis of triacylglycerols to glycerol and fatty-acid salts. A fatty-acid carboxylate has a polar ionic head and a nonpolar hydrocarbon tail, allowing micelle formation under suitable conditions.

Visual card: Saponification of Triacylglycerols, Irreversible Acid–Base Driving Force & Micelles

Chemistry diagram: Aggregate cartoon: each head/tail token represents an amphiphile, not atoms or a resolved molecular structure.

Triacylglycerol + 3 NaOH → Glycerol + 3 $RCOO^-Na^+$ (Soap)

Micelle: Hydrophobic Tails In, $–COO^-$ Heads Out

Saponification (Ester + NaOH) | Irreversible in base! | Step 1: HO^- attacks C=O → Step 2: RO^- leaves → Step 3: RO^- deprotonates RCOOH

Soap Micelle in Water | Amphiphilic fatty acid salt (C_{12}–C_{18}) | Nonpolar tails dissolve grease inside core; ionic $–COO^-$ heads face H_2O

Hard Water (Ca^{2+} / Mg^{2+}) | Forms insoluble $(RCOO)_2Ca$↓ scum | Sulfonate detergents ($R–SO_3^-Na^+$) stay soluble in hard water

Step 1–2: Nucleophilic Addition of HO^- & Alkoxide Expulsion — — Hydroxide attacks the ester carbonyl to form a tetrahedral intermediate, which collapses by expelling the alkoxide (RO^-) to form a carboxylic acid (RCOOH) andRO^-.

Step 3: The Irreversible Acid–Base Driving Force! — — Alkoxide (RO^-, conjugate acid pKa ~16) immediately deprotonates the carboxylic acid (RCOOH, pKa ~5) to form a resonance-stabilized carboxylate ($RCOO^-$, ΔpKa > 11 → K_eq > 10^{11})!

Amphiphilic Self-Assembly (Micelles vs. Bilayers) — — Single-tailed fatty acid salts (cone-shaped) pack into spherical MICELLES above the critical micelle concentration (CMC); double-tailed phospholipids (cylindrical) form planar BILAYERS.

soap:cue | Memory Cues

Head interacts with water; tails gather away from water.

Visual card: Why Saponification Is Irreversible & How Micelles Solubilize Oils

What aspect of ester hydrolysis or lipid assembly is tested?

Why Is Base-Promoted Ester Hydrolysis Irreversible? — Final Deprotonation (RCOOH + RO^- → $RCOO^-$ + ROH) Locks the Product! — Once the negatively charged carboxylate ($RCOO^-$) forms, nucleophilic attack by ROH is electrostatically impossible; NaOH is a stoichiometric reagent, not a catalyst!

Single-Chain Soap vs. Double-Chain Phospholipid — 1 Chain (Cone) = Spherical Micelle; 2 Chains (Cylinder) = Bilayer — Steric packing geometry dictates whether an amphiphile forms a single-layer micelle core or a two-leaflet cell membrane.

What Drives Micelle Formation Thermodynamically? — Hydrophobic Effect: ↑ Entropy (ΔS > 0) of Released Bulk Water Molecules! — Burying nonpolar hydrocarbon tails releases ordered "clathrate cage" water molecules back into bulk water, increasing solvent entropy.

soap:trap | Common Mistakes / MCAT Traps

Neutralizing a free fatty acid makes its salt, but that acid–base reaction is not ester hydrolysis. A micelle has hydrophobic material inside; it is not the same structure as a lipid bilayer.

Visual card: Saying Hydrophobic Tails "Attract Each Other Strongly" as the Primary Driver of Micelles

Common Trap — — Claiming micelles form primarily because nonpolar tails have stronger attractions than water-water H-bonds.

Correct Reasoning — — The hydrophobic effect is ENTROPY-driven (ΔS_water > 0): water molecules maximize their hydrogen-bonding freedom by excluding nonpolar chains into a shared core!

soap:app | MCAT Applications

Soap can disperse nonpolar material into water through aggregates. Aggregation depends on concentration, ionic conditions, and tail structure.

Visual card: Bile Salt Emulsification, Pancreatic Lipase & Fatty Acid Melting Points

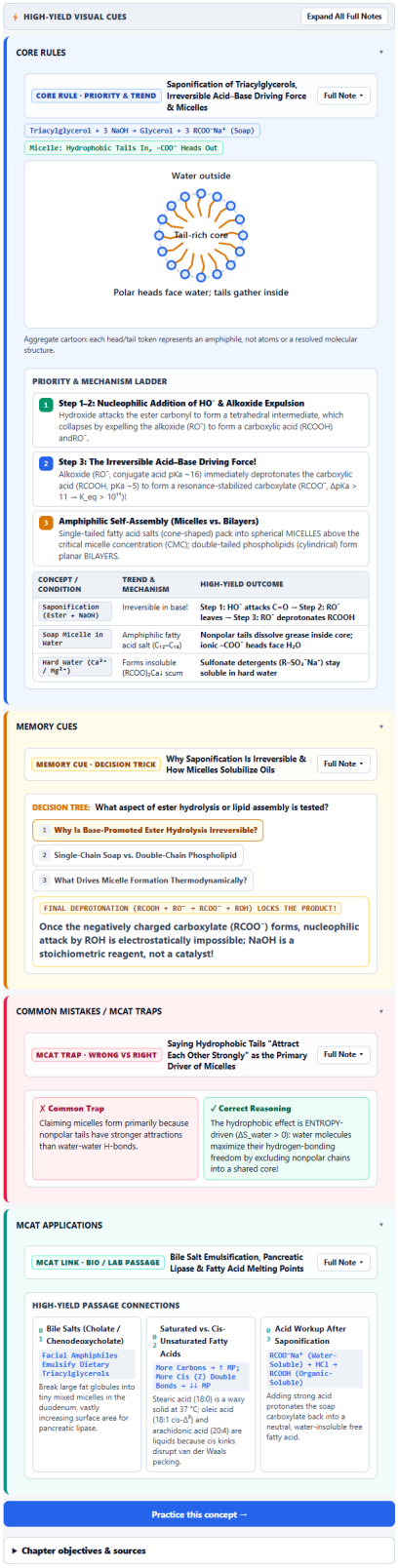
Chemistry diagram: Aggregate cartoon: each head/tail token represents an amphiphile, not atoms or a resolved molecular structure.

Bile Salts (Cholate / Chenodeoxycholate) — Facial Amphiphiles Emulsify Dietary Triacylglycerols — Break large fat globules into tiny mixed micelles in the duodenum, vastly increasing surface area for pancreatic lipase.

Saturated vs. Cis-Unsaturated Fatty Acids — More Carbons → ↑ MP; More Cis (Z) Double Bonds → ↓↓ MP — Stearic acid (18:0) is a waxy solid at 37 °C; oleic acid (18:1 cis-Δ^9) and arachidonic acid (20:4) are liquids because cis kinks disrupt van der Waals packing.

Acid Workup After Saponification — $RCOO^-Na^+$ (Water-Soluble) + HCl → RCOOH (Organic-Soluble) — Adding strong acid protonates the soap carboxylate back into a neutral, water-insoluble free fatty acid.

Screenshot: soap-visuals



soap:step-1 | 1. Find the head group

Laurate is a long-chain carboxylate. Its negative charge is represented on one O but delocalized over both. Counterions are omitted from this molecular-ion graph.

Displayed molecule ID: laurate

{"highlights": {"prin": [0, 12, 13]}, "revealName": true}

Screenshot: soap-step-1



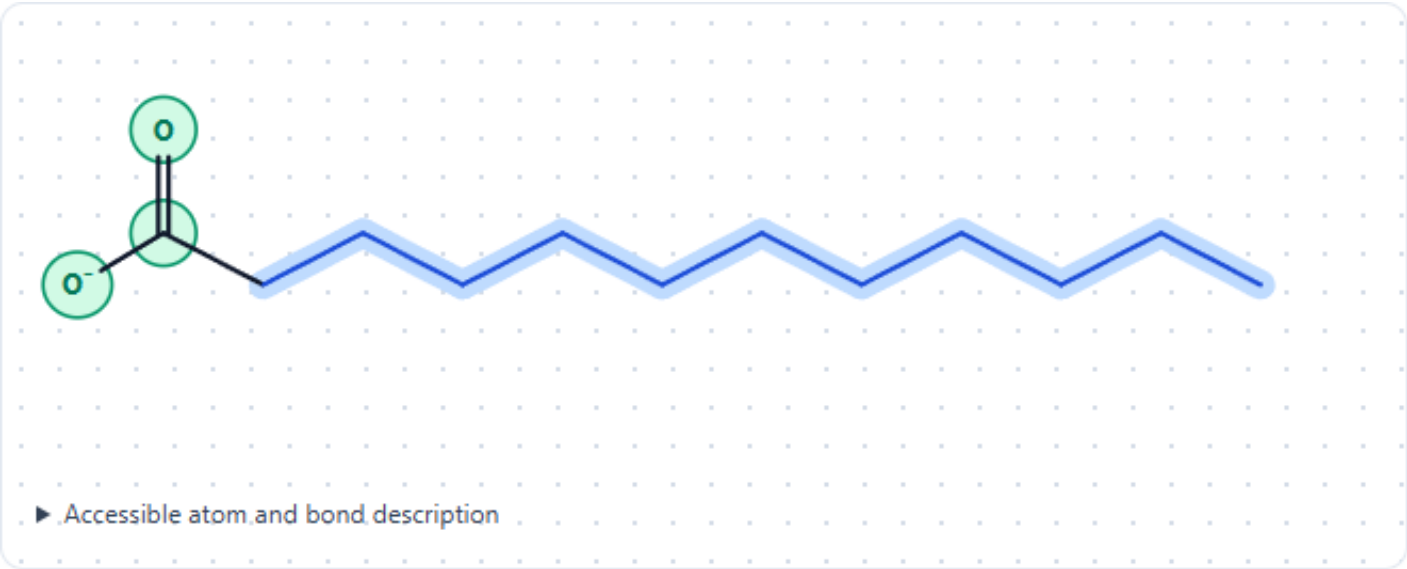
soap:step-2 | 2. Contrast the tail

The hydrocarbon tail interacts favorably with other nonpolar material. Many such tails can cluster while ionic heads contact the surrounding water.

Displayed molecule ID: laurate

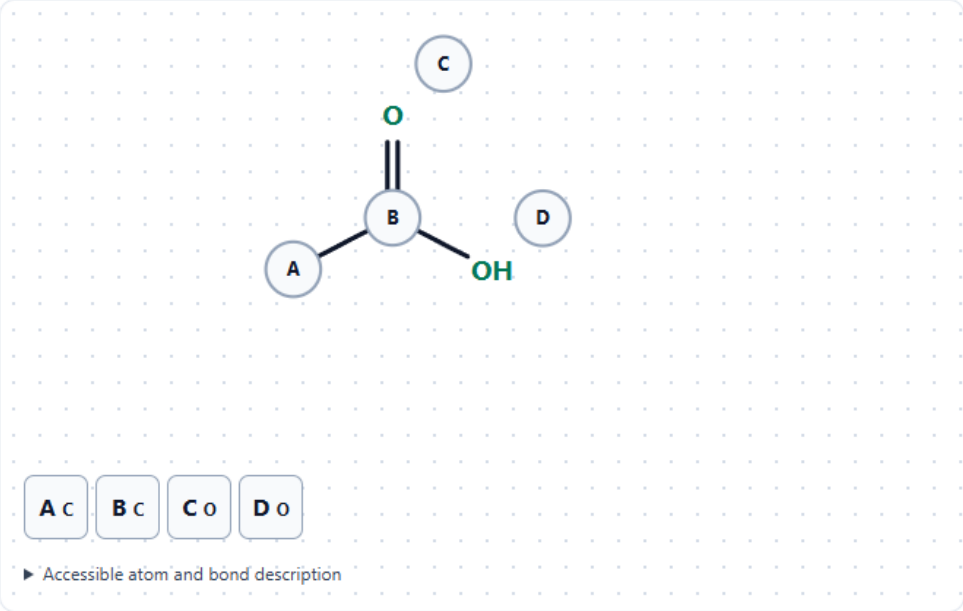
{"highlights": {"path": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11], "prin": [0, 12, 13]}, "revealName": false}

Screenshot: soap-step-2



acid-ratio | Calculate acid ionization

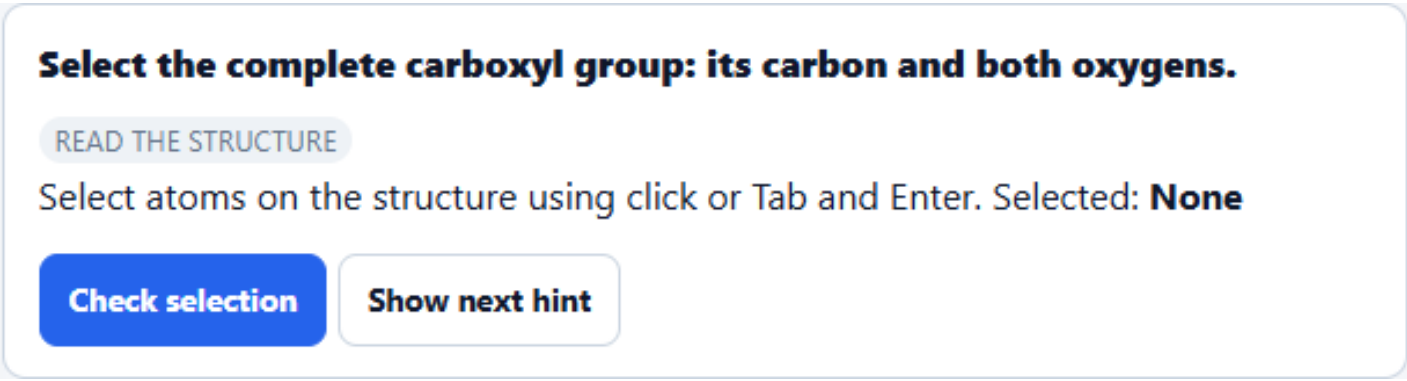
Screenshot: acid-ratio-stage-1-molecule



acid-ratio:stage-1 | Select the complete carboxyl group: its carbon and both oxygens.

Answer type: atoms

Screenshot: acid-ratio-stage-1-question



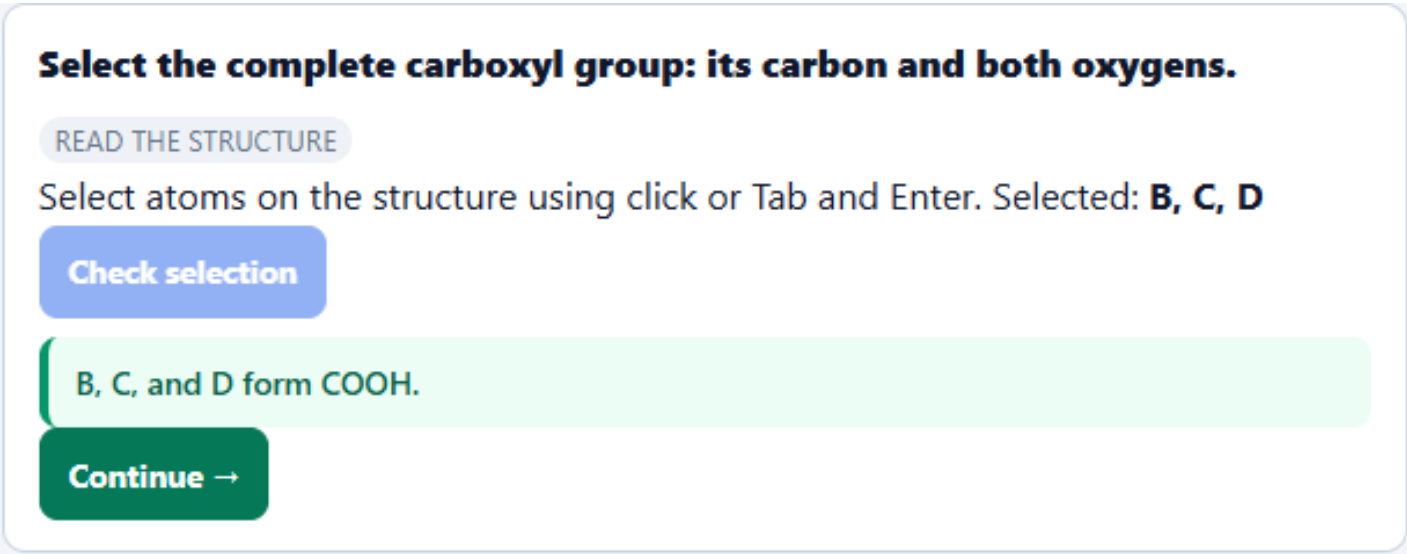
{"authorAnswer": [1, 2, 3], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2, 3]}}

Hint 1: Include the shared carbon.

Hint 2: The methyl carbon is outside the group.

Explanation: B, C, and D form COOH.

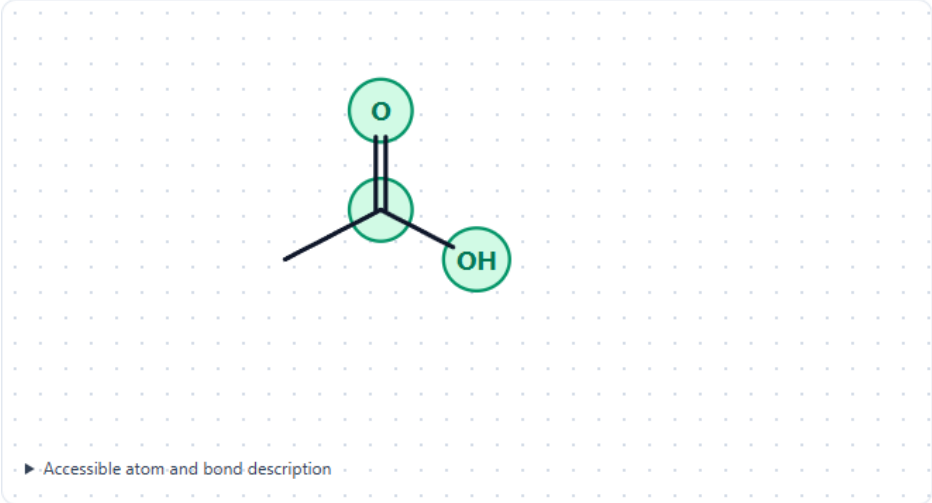
Screenshot: acid-ratio-stage-1-feedback



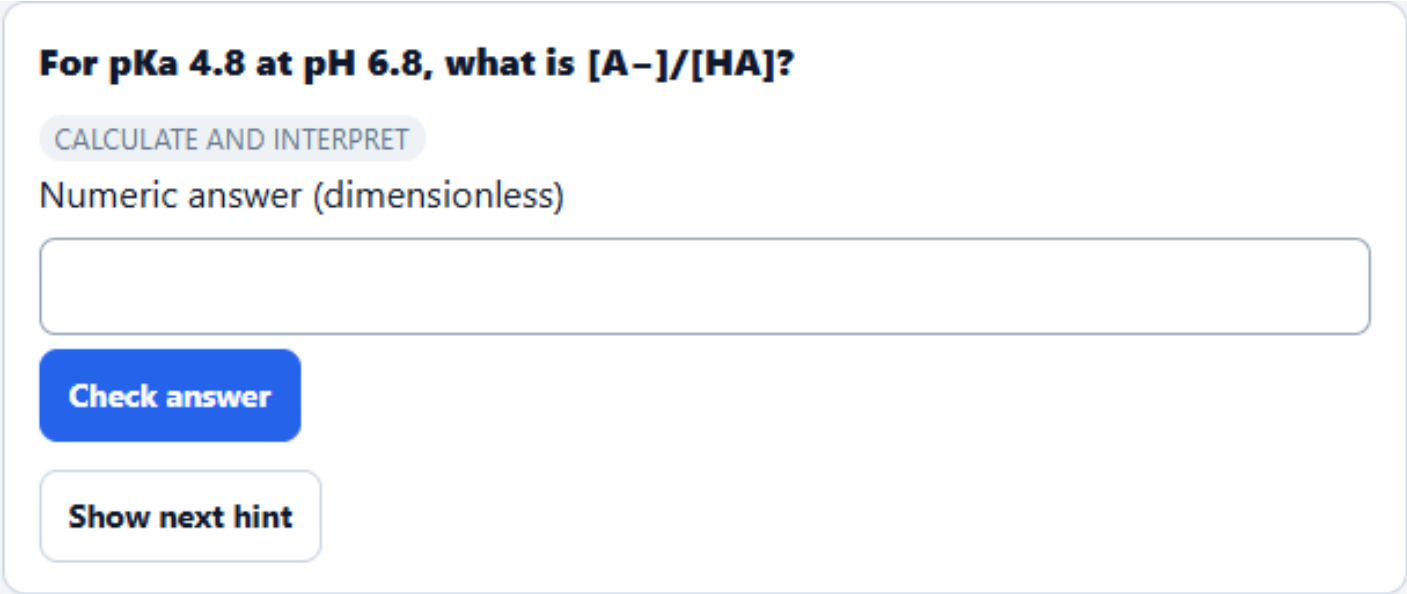
acid-ratio:stage-2 | For pKa 4.8 at pH 6.8, what is [A−]/[HA]?

Answer type: number | unit: dimensionless

Screenshot: acid-ratio-stage-2-molecule



Screenshot: acid-ratio-stage-2-question



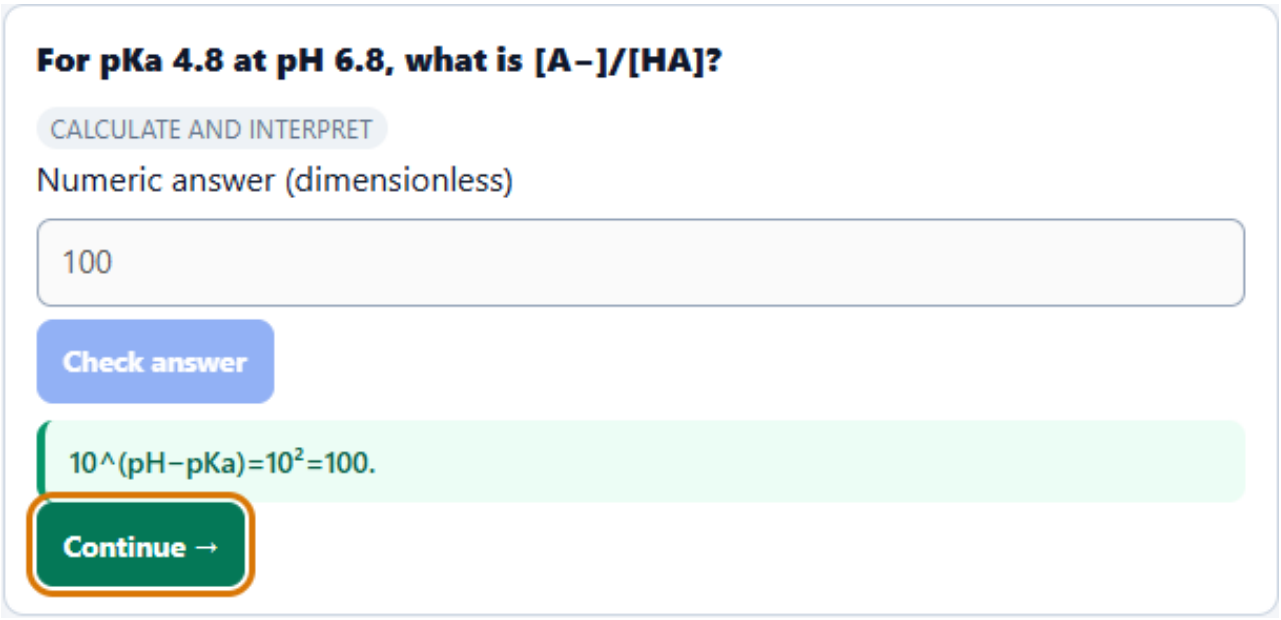
{"authorAnswer": 100, "tolerance": 0, "highlightsAfterCorrectAnswer": {"prin": [1, 2, 3]}}

Hint 1: Use Henderson–Hasselbalch.

Hint 2: The pH exceeds pKa by two.

Explanation: 10^(pH−pKa)=10^2=100.

Screenshot: acid-ratio-stage-2-feedback



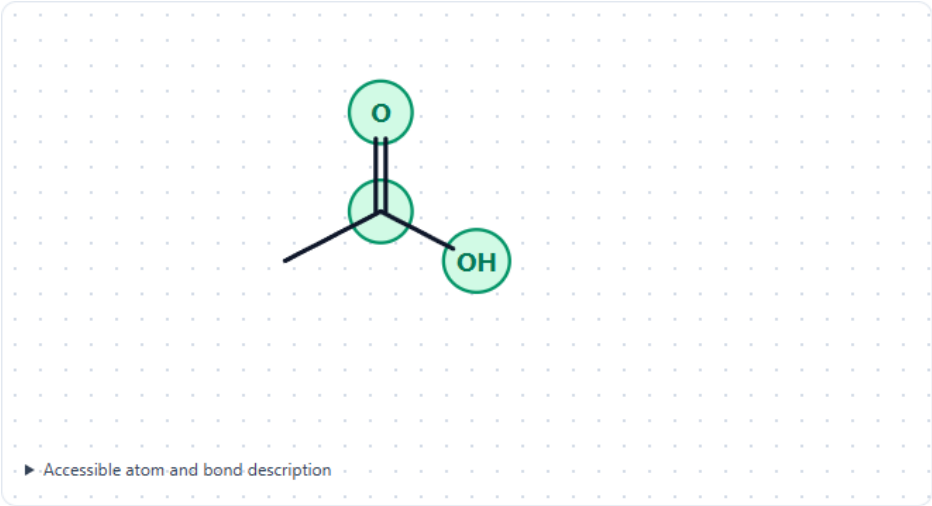
acid-ratio:stage-3 | Which form predominates?

Answer type: choice

Option 0 (A): Neutral acid

Option 1 (B): Carboxylate

Screenshot: acid-ratio-stage-3-molecule



Screenshot: acid-ratio-stage-3-question



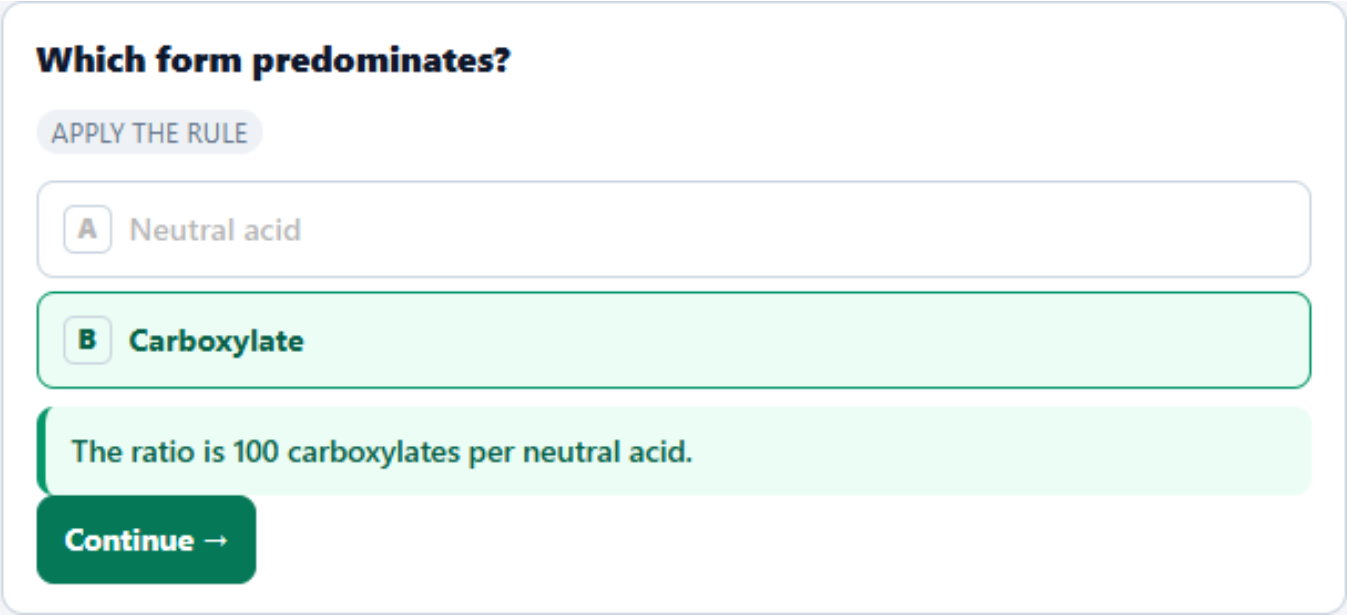
{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2, 3]}}

Hint 1: Use the calculated ratio.

Hint 2: Higher pH favors deprotonation.

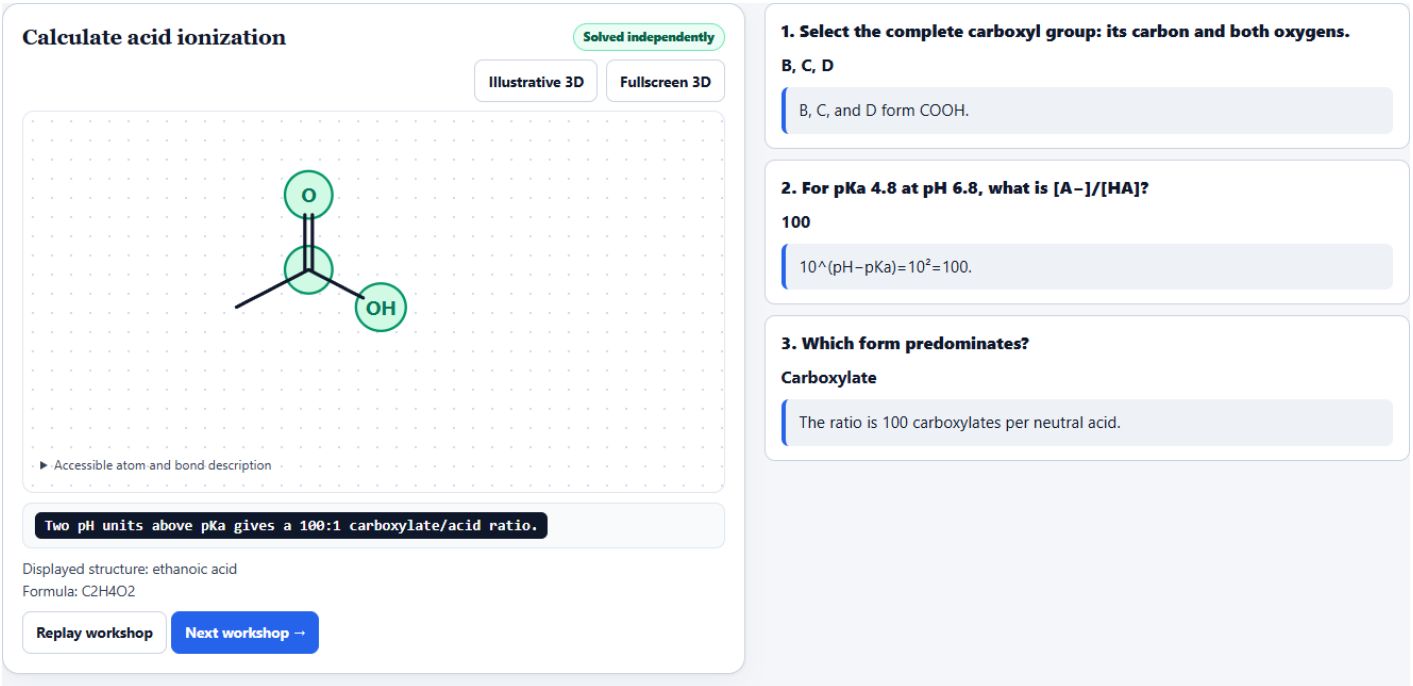
Explanation: The ratio is 100 carboxylates per neutral acid.

Screenshot: acid-ratio-stage-3-feedback



Completion summary: Two pH units above pKa gives a 100:1 carboxylate/acid ratio.

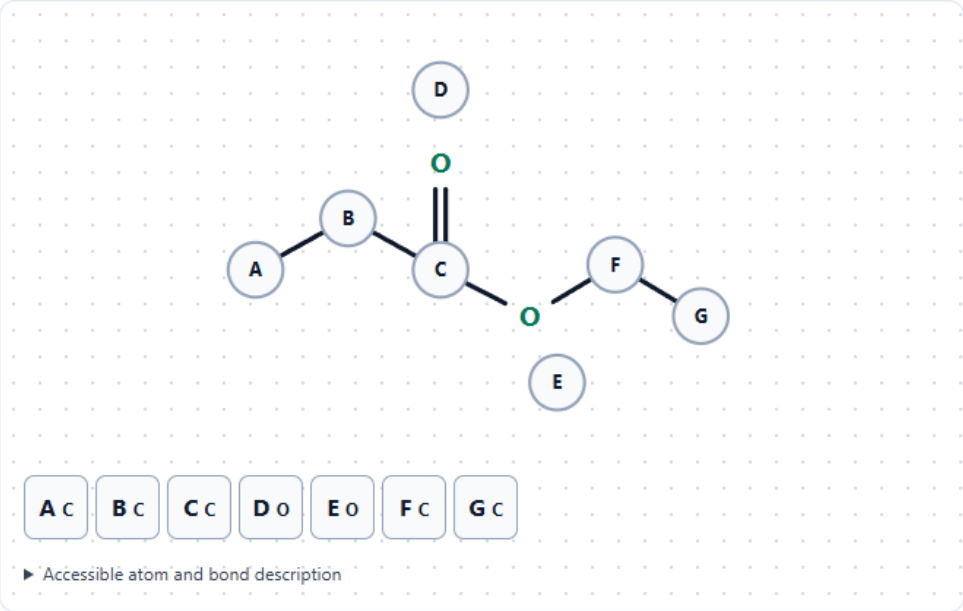
Screenshot: acid-ratio-complete



make-ester | Reason about acid derivatives

The ester canvas illustrates the functional-group pattern. The later acid/amine question is a separate transfer question, not a transformation of this ester.

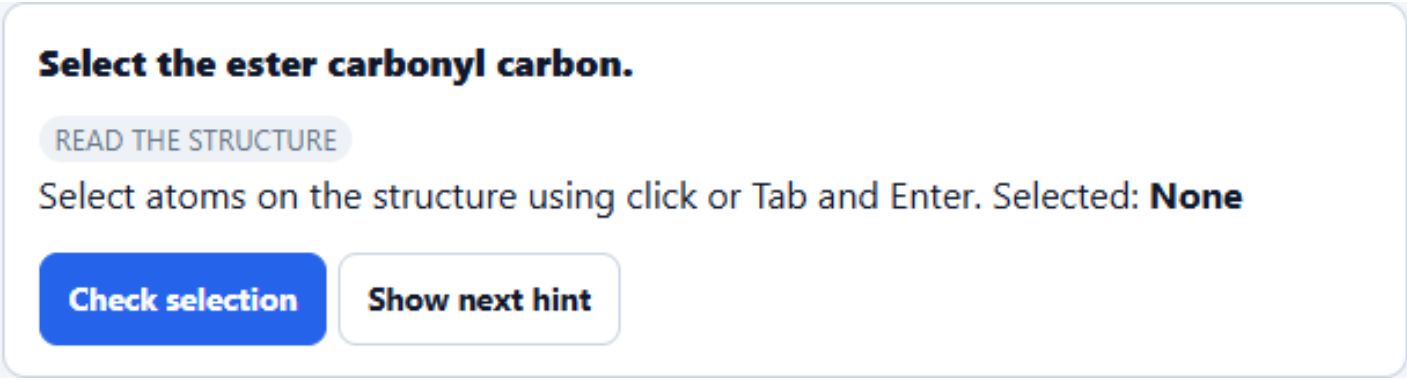
Screenshot: make-ester-stage-1-molecule



make-ester:stage-1 | Select the ester carbonyl carbon.

Answer type: atoms

Screenshot: make-ester-stage-1-question



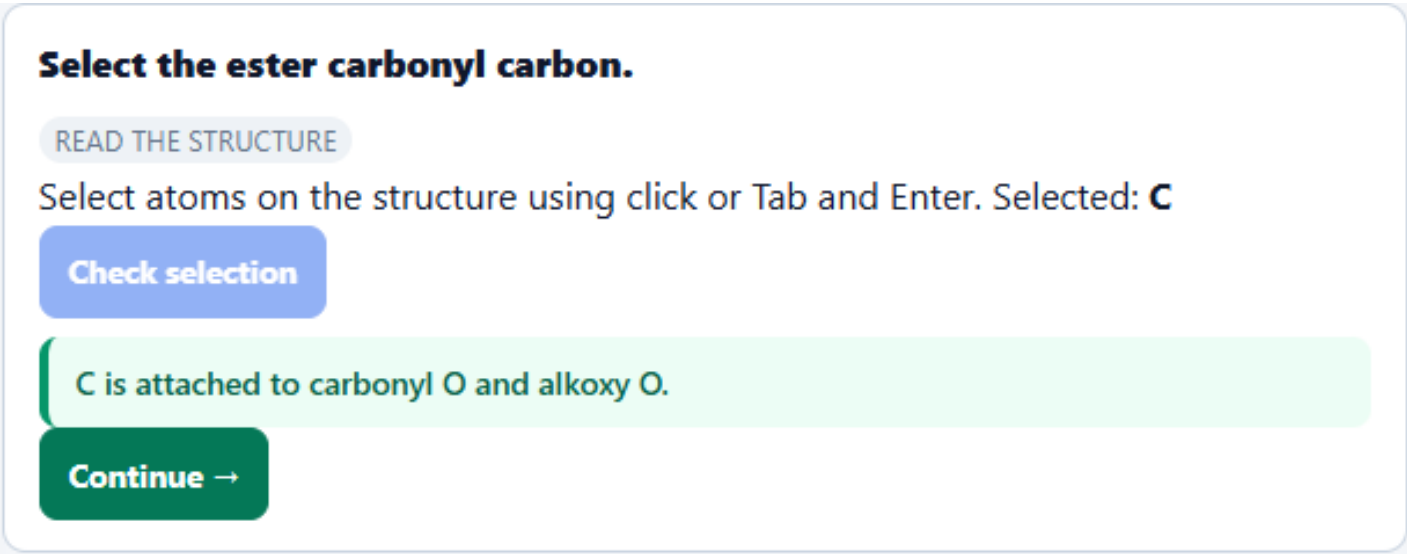
{"authorAnswer": [2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [2, 3, 4]}}

Hint 1: Locate C(=O)–O–C.

Hint 2: Choose the shared carbon.

Explanation: C is attached to carbonyl O and alkoxy O.

Screenshot: make-ester-stage-1-feedback



make-ester:stage-2 | Removing water from a Fischer esterification generally shifts toward...

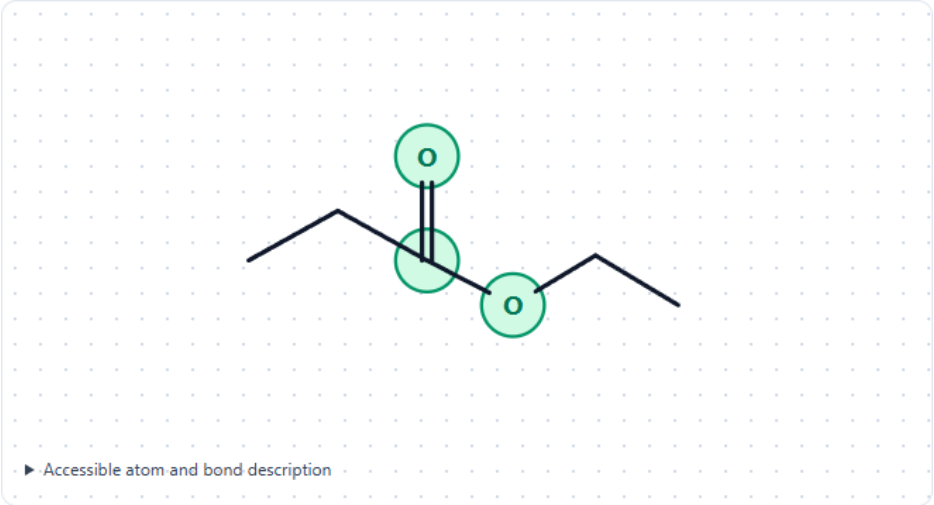
Answer type: choice

Option 0 (A): Ester

Option 1 (B): Acid and alcohol

Option 2 (C): No equilibrium change

Screenshot: make-ester-stage-2-molecule



Screenshot: make-ester-stage-2-question



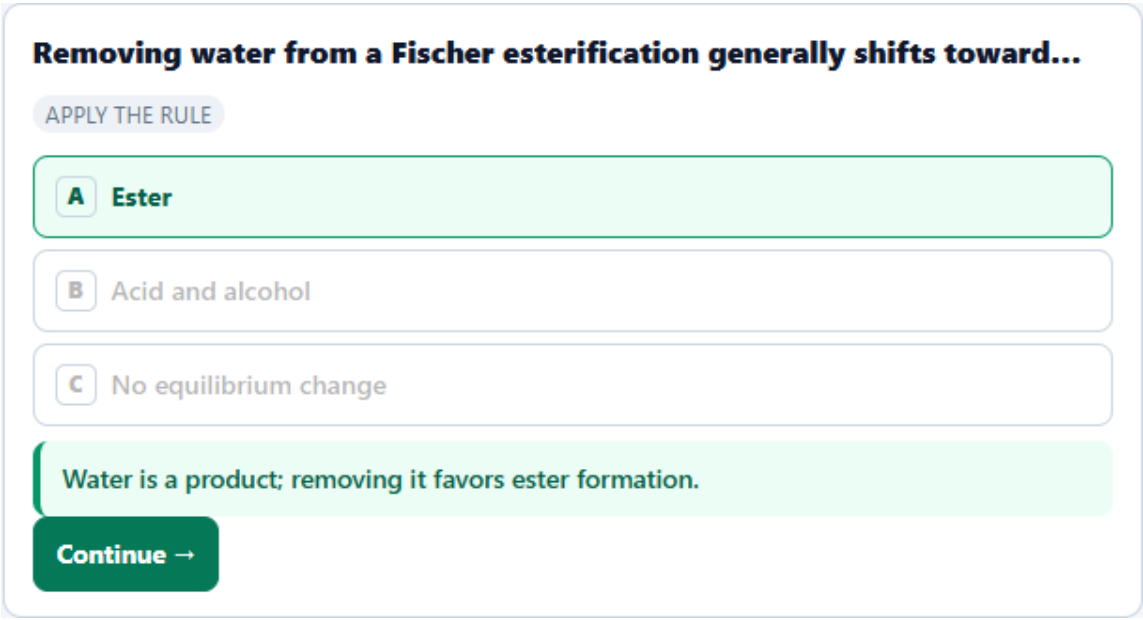
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [2, 3, 4]}}

Hint 1: Write acid + alcohol ⇌ ester + water.

Hint 2: Apply Le Châtelier's principle.

Explanation: Water is a product; removing it favors ester formation.

Screenshot: make-ester-stage-2-feedback



make-ester:stage-3 | Mixing a carboxylic acid and an amine in water initially favors...

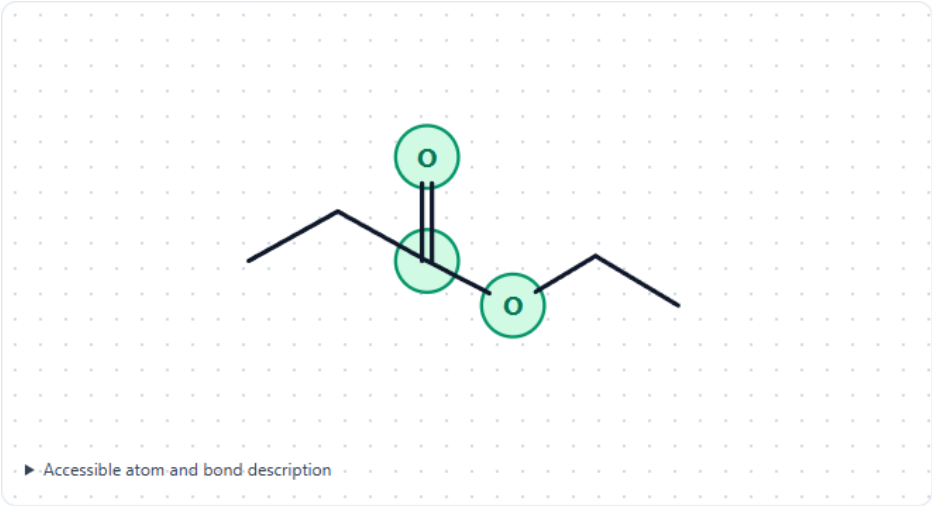
Answer type: choice

Option 0 (A): Ammonium carboxylate salt

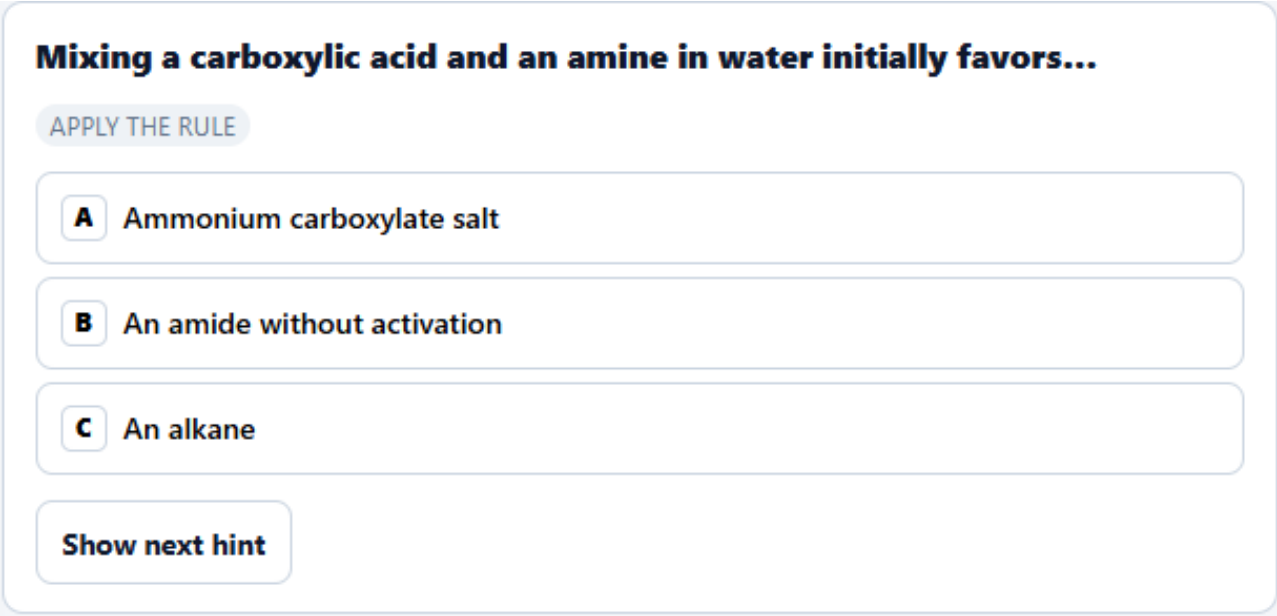
Option 1 (B): An amide without activation

Option 2 (C): An alkane

Screenshot: make-ester-stage-3-molecule



Screenshot: make-ester-stage-3-question



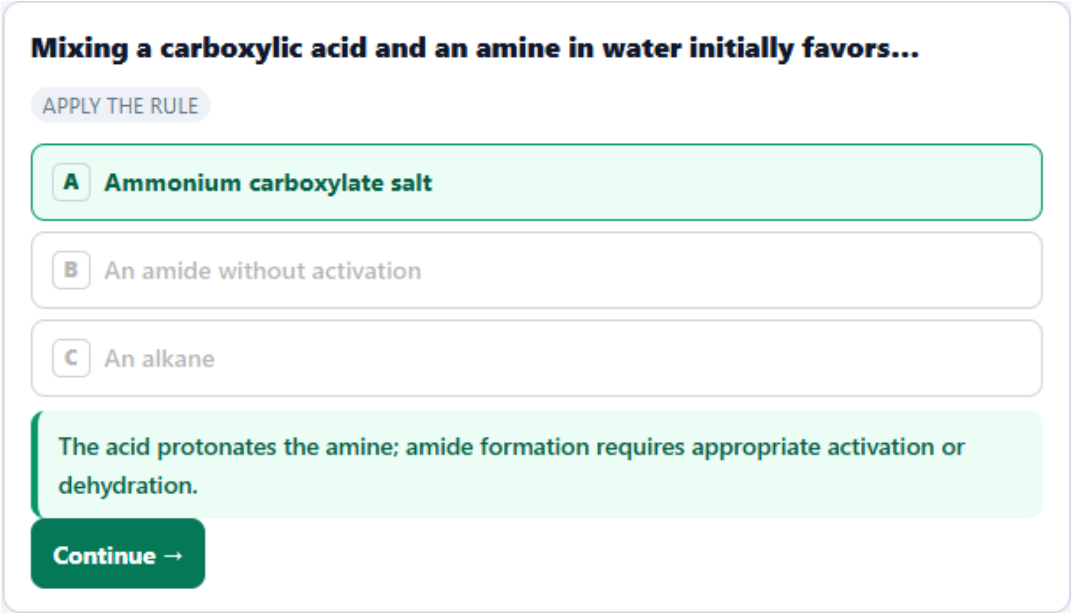
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Consider the fast acid–base step.

Hint 2: The amine is a base.

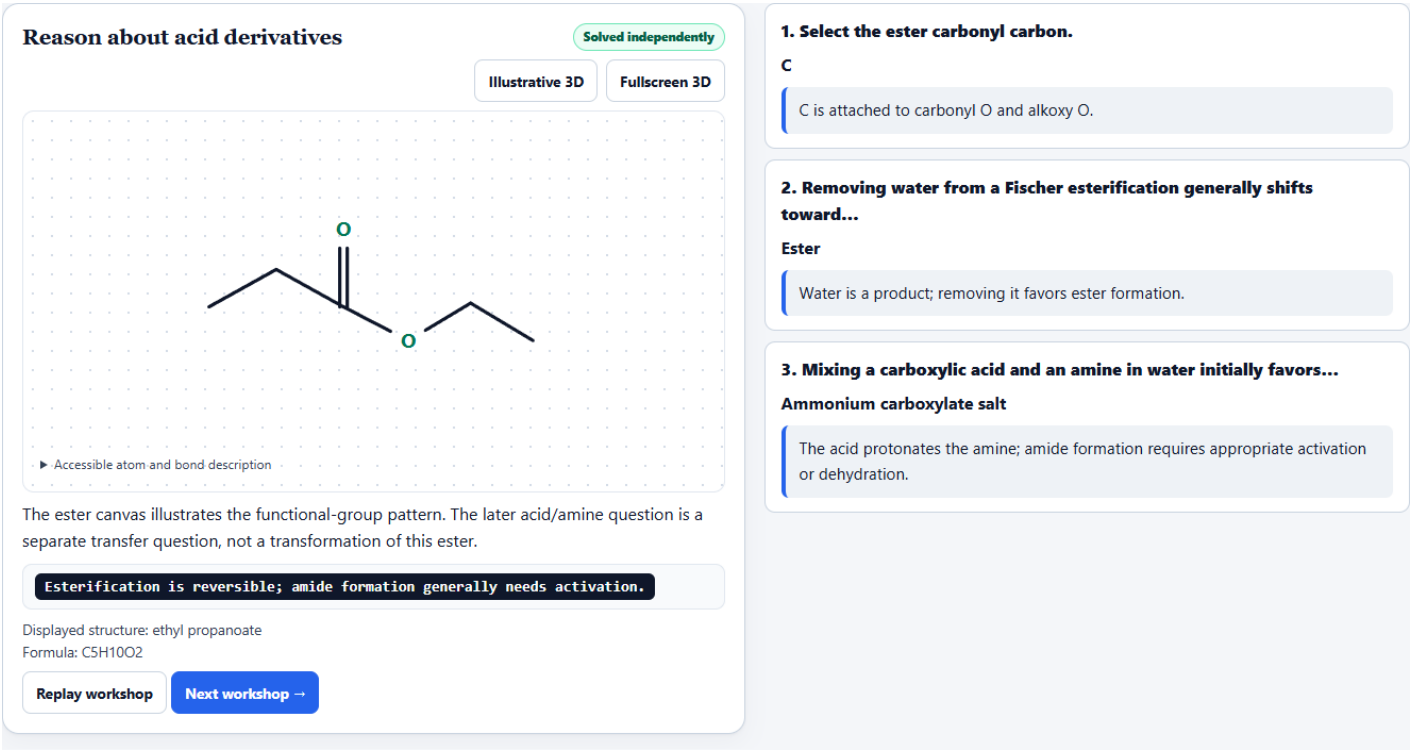
Explanation: The acid protonates the amine; amide formation requires appropriate activation or dehydration.

Screenshot: make-ester-stage-3-feedback



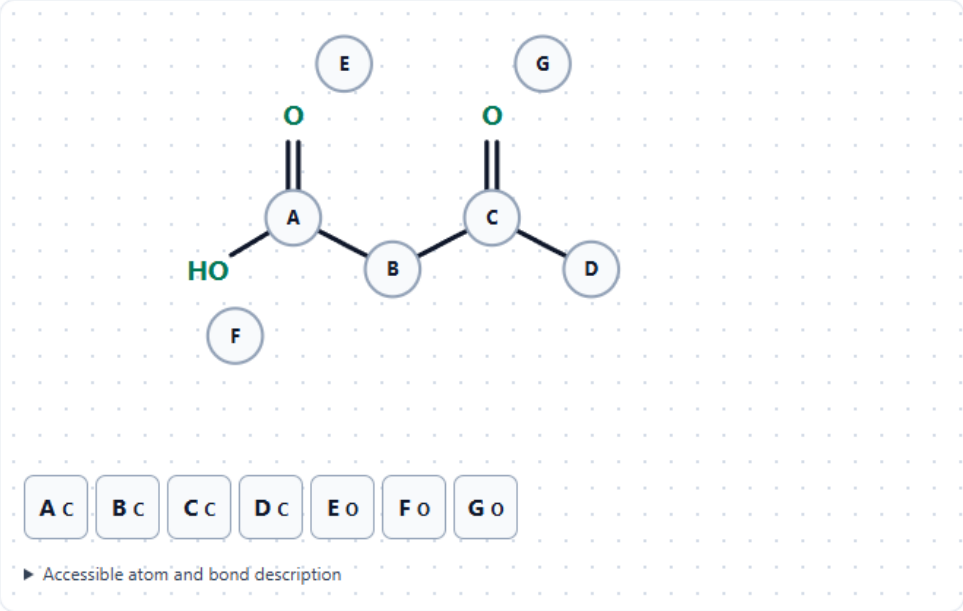
Completion summary: Esterification is reversible; amide formation generally needs activation.

Screenshot: make-ester-complete



lose-co2 | Track decarboxylation carbons

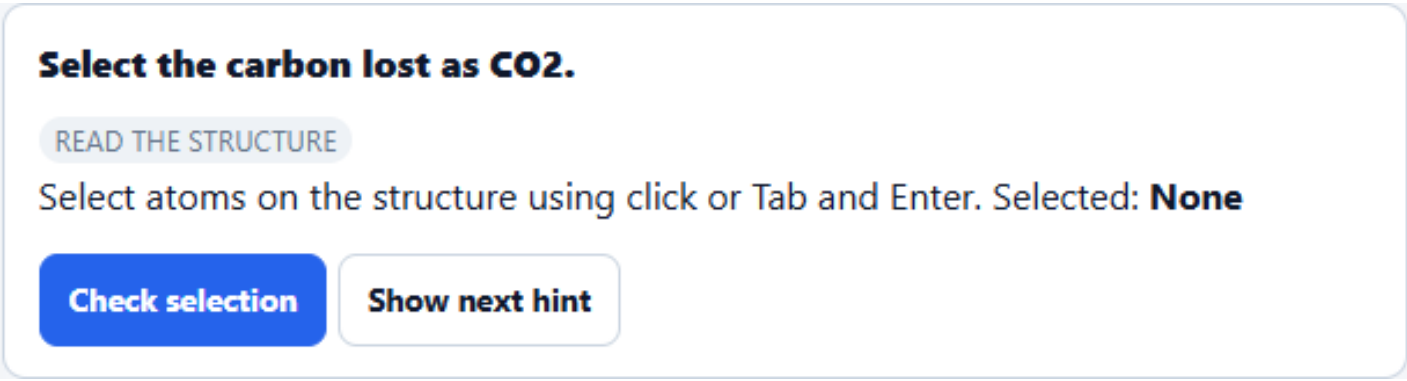
Screenshot: lose-co2-stage-1-molecule



lose-co2:stage-1 | Select the carbon lost as CO2.

Answer type: atoms

Screenshot: lose-co2-stage-1-question



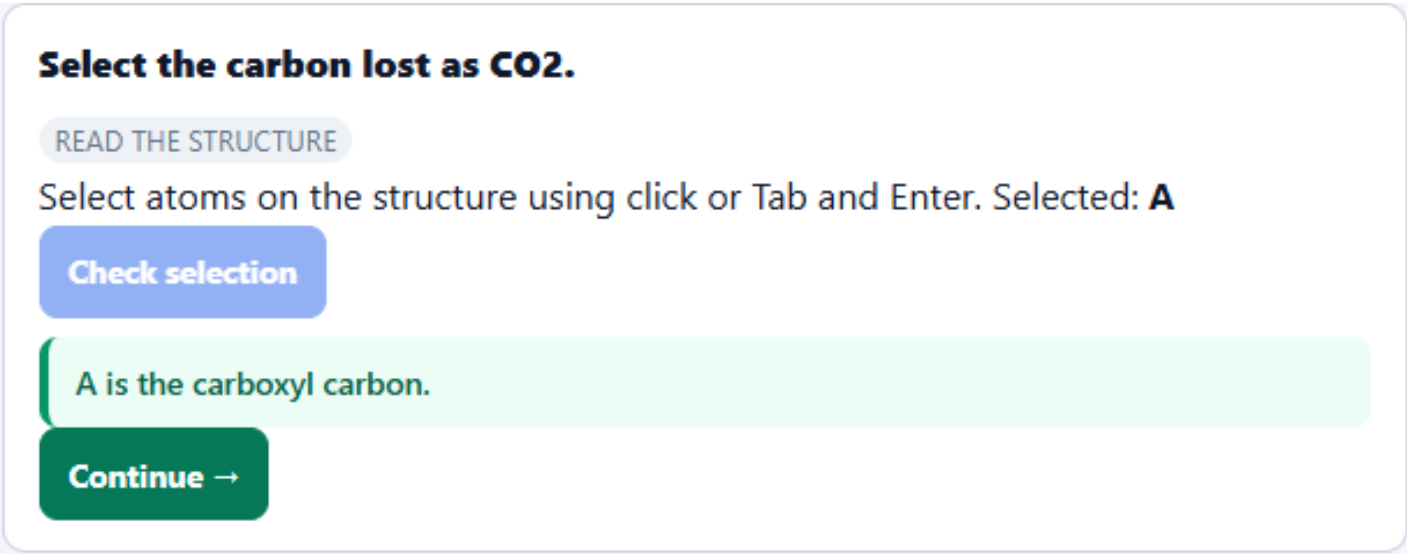
{"authorAnswer": [0], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0, 4, 5]}}

Hint 1: The carboxyl group is expelled.

Hint 2: Do not choose the ketone carbon.

Explanation: A is the carboxyl carbon.

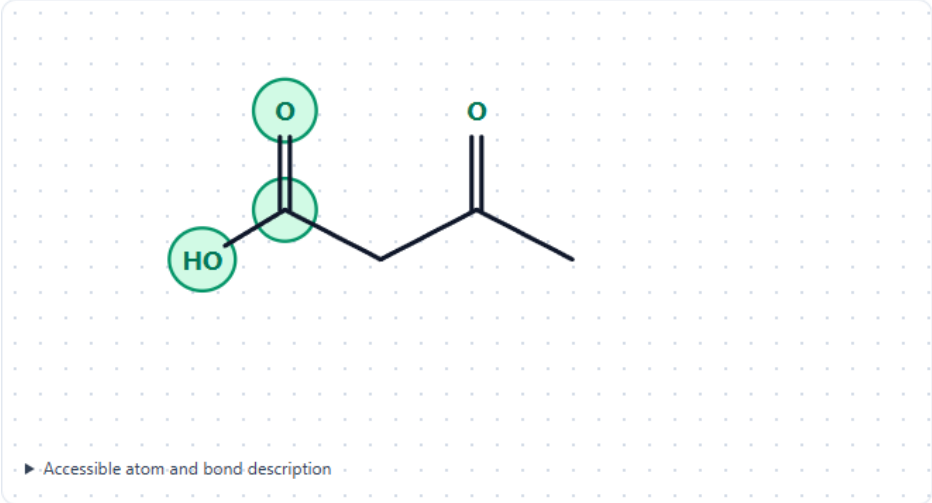
Screenshot: lose-co2-stage-1-feedback



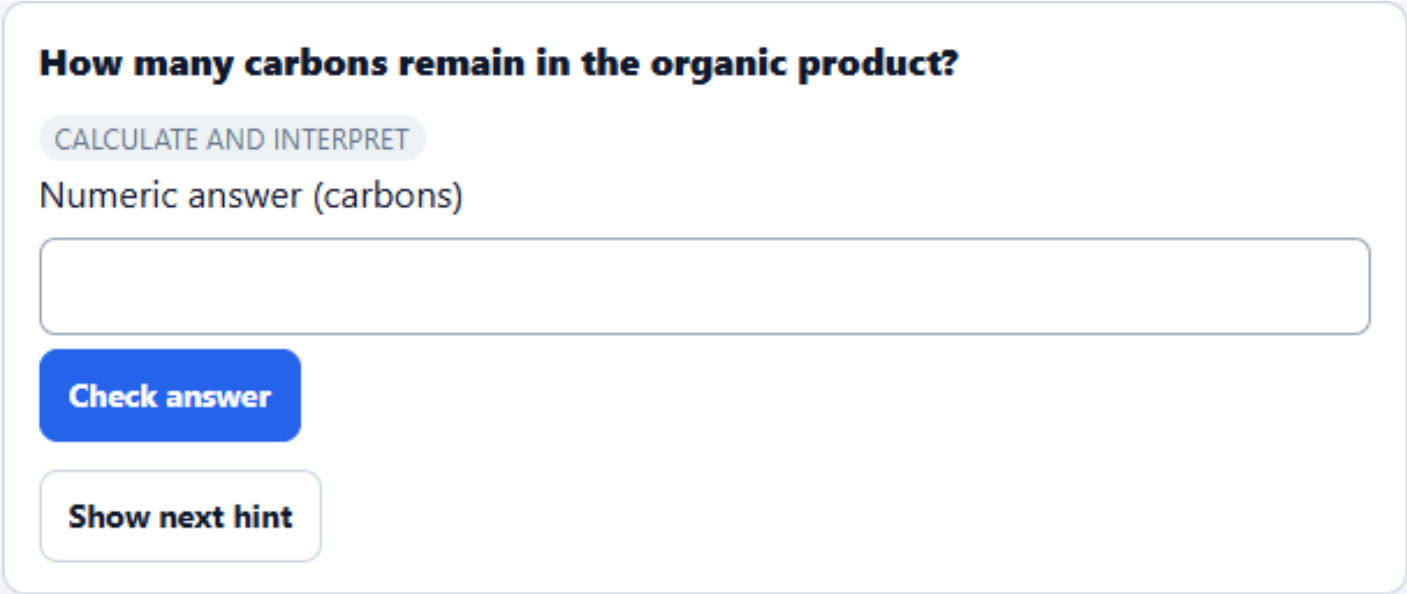
lose-co2:stage-2 | How many carbons remain in the organic product?

Answer type: number | unit: carbons

Screenshot: lose-co2-stage-2-molecule



Screenshot: lose-co2-stage-2-question



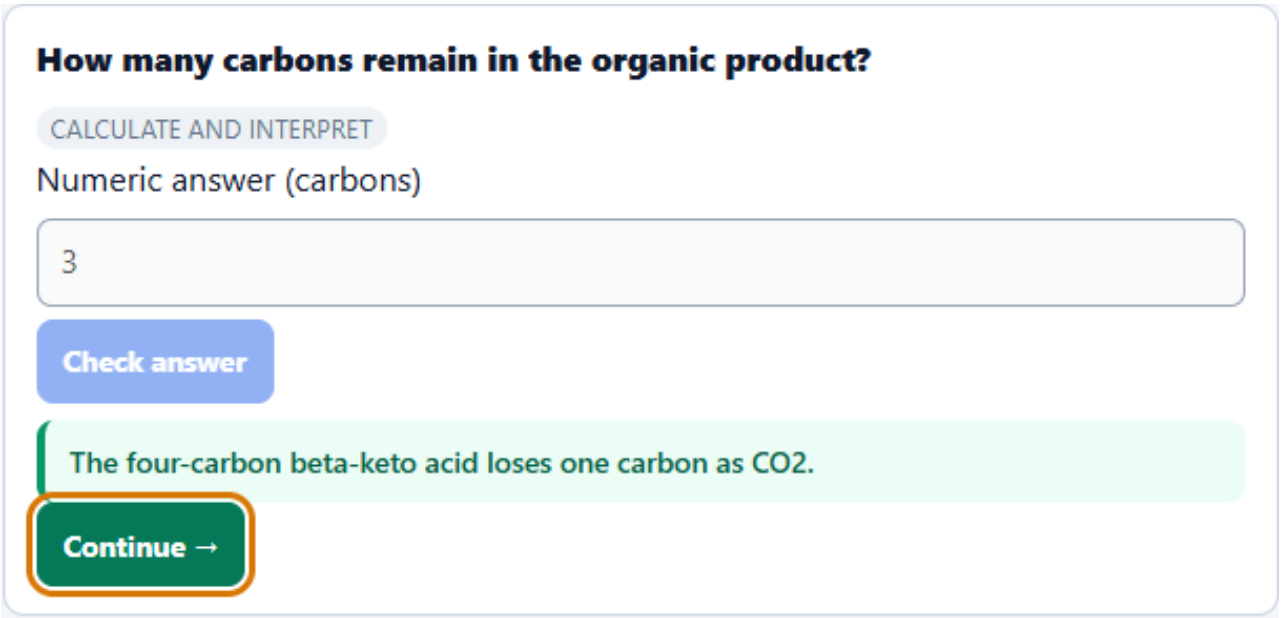
{"authorAnswer": 3, "tolerance": 0, "highlightsAfterCorrectAnswer": {"prin": [0, 4, 5]}}

Hint 1: Subtract the departing carboxyl carbon.

Hint 2: 4 − 1 = ?

Explanation: The four-carbon beta-keto acid loses one carbon as CO2.

Screenshot: lose-co2-stage-2-feedback

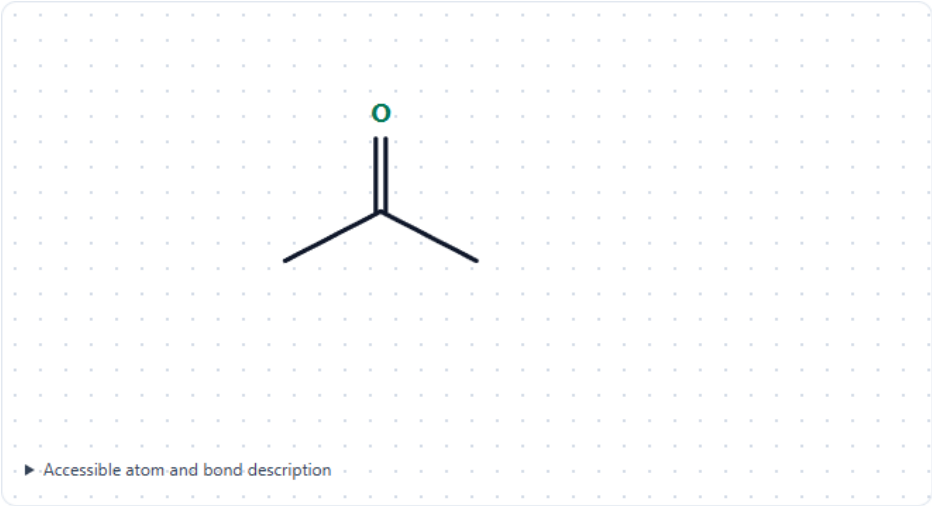


This stage uses a different molecule (connectivity shown).

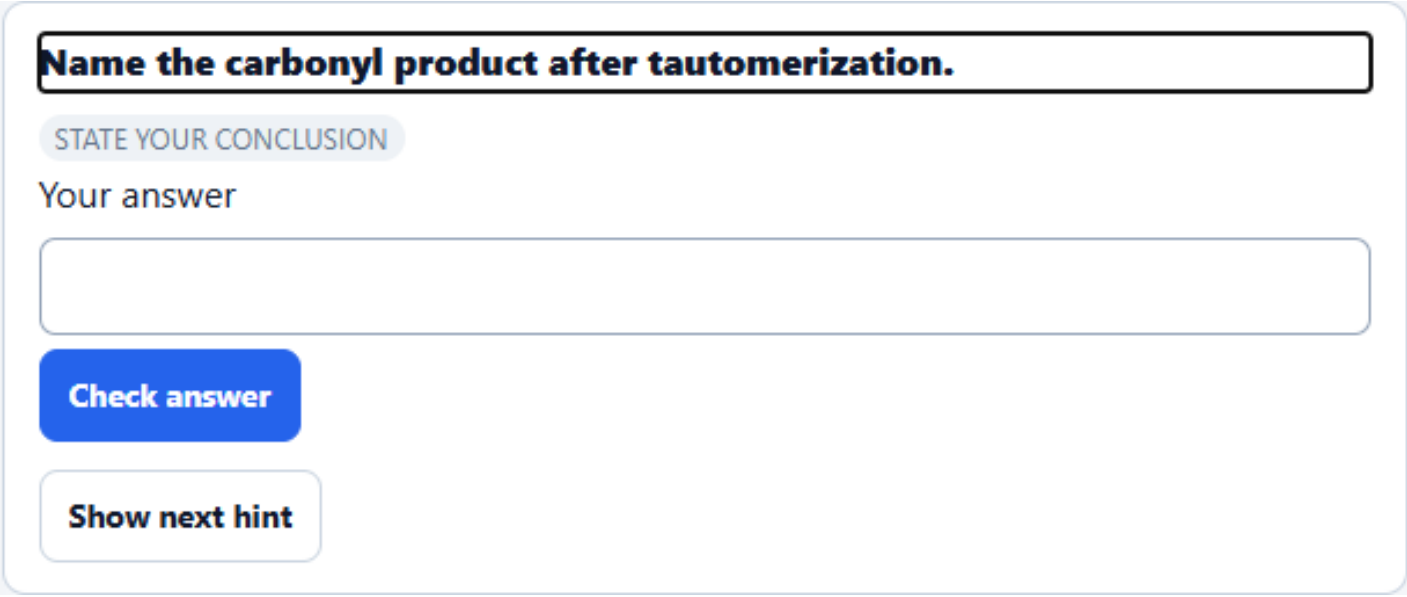
lose-co2:stage-3 | Name the carbonyl product after tautomerization.

Answer type: text

Screenshot: lose-co2-stage-3-molecule



Screenshot: lose-co2-stage-3-question



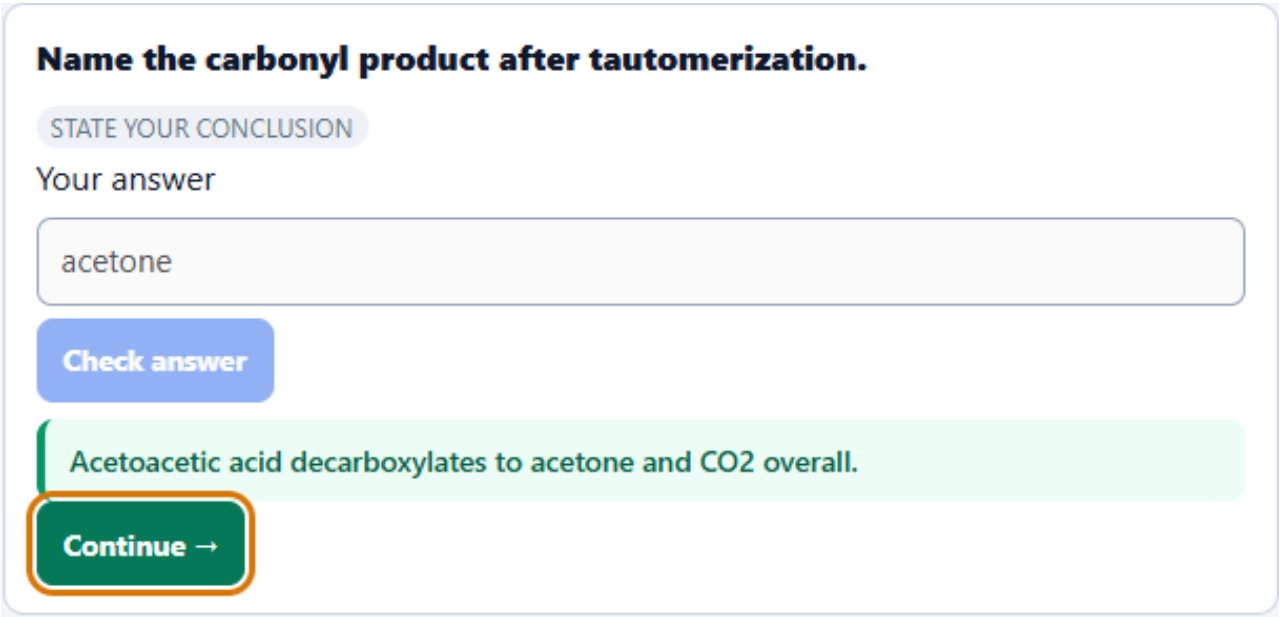
{"authorAnswer": ["acetone", "propanone", "propan-2-one", "2-propanone", "dimethyl ketone"], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 3]}}

Hint 1: The remaining chain has three carbons.

Hint 2: The carbonyl becomes the middle carbon of that chain.

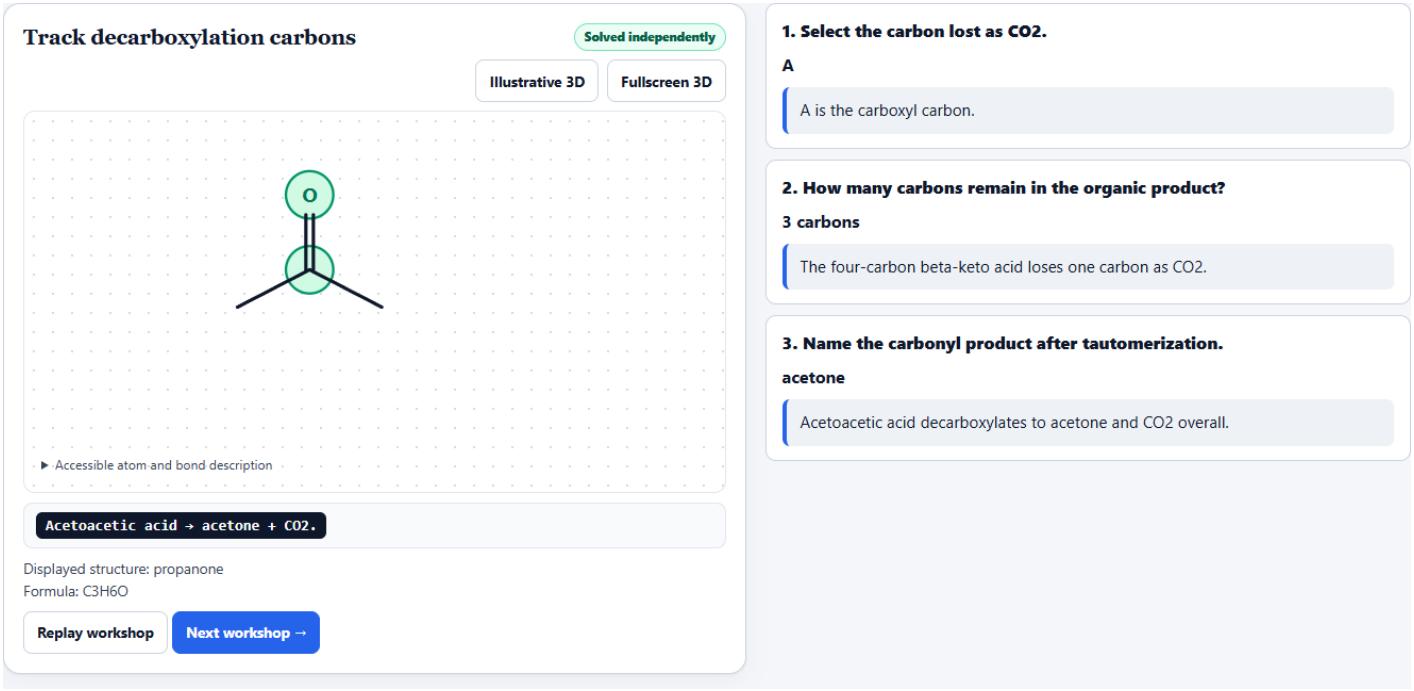
Explanation: Acetoacetic acid decarboxylates to acetone and CO2 overall.

Screenshot: lose-co2-stage-3-feedback



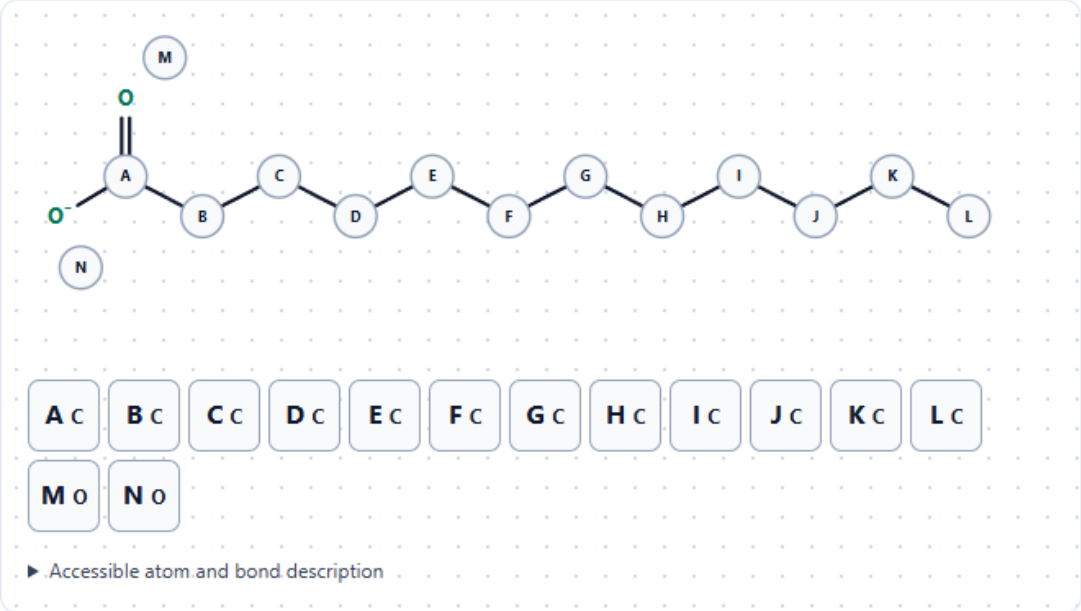
Completion summary: Acetoacetic acid → acetone + CO2.

Screenshot: lose-co2-complete



soap-head | Connect structure and aggregation

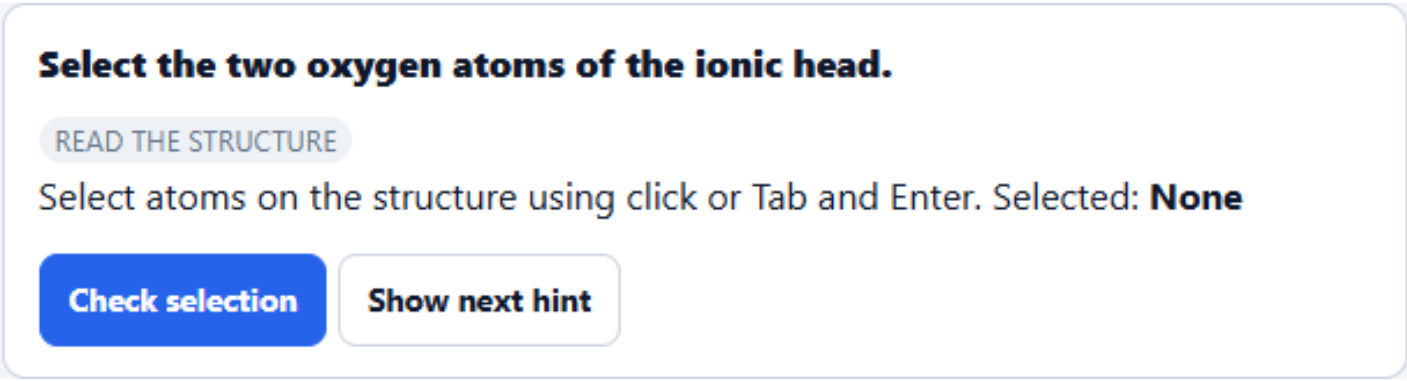
Screenshot: soap-head-stage-1-molecule



soap-head:stage-1 | Select the two oxygen atoms of the ionic head.

Answer type: atoms

Screenshot: soap-head-stage-1-question



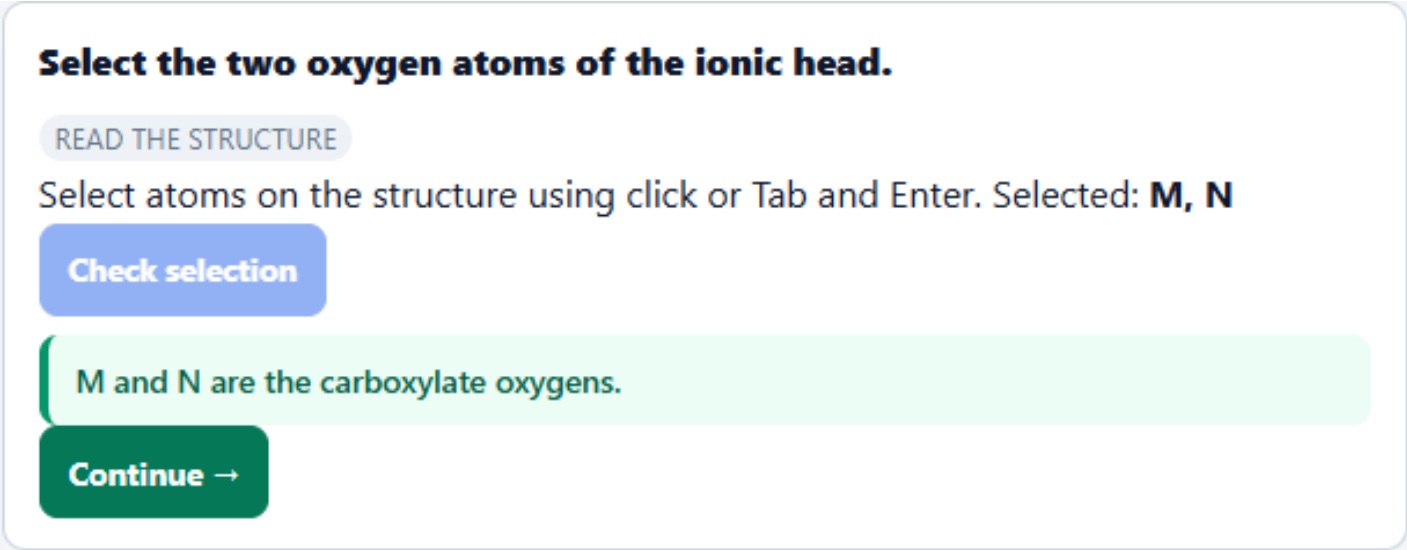
{"authorAnswer": [12, 13], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [12, 13]}}

Hint 1: Locate the O labels at the end of the chain.

Hint 2: The charge is delocalized across the two oxygens.

Explanation: M and N are the carboxylate oxygens.

Screenshot: soap-head-stage-1-feedback



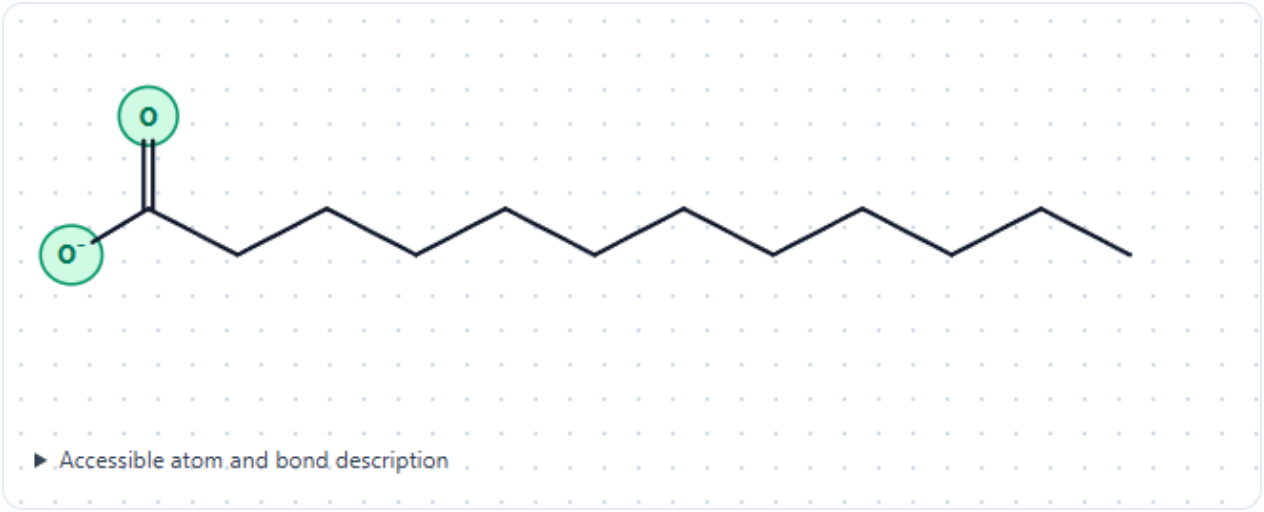
soap-head:stage-2 | In an aqueous micelle, hydrocarbon tails tend to point…

Answer type: choice

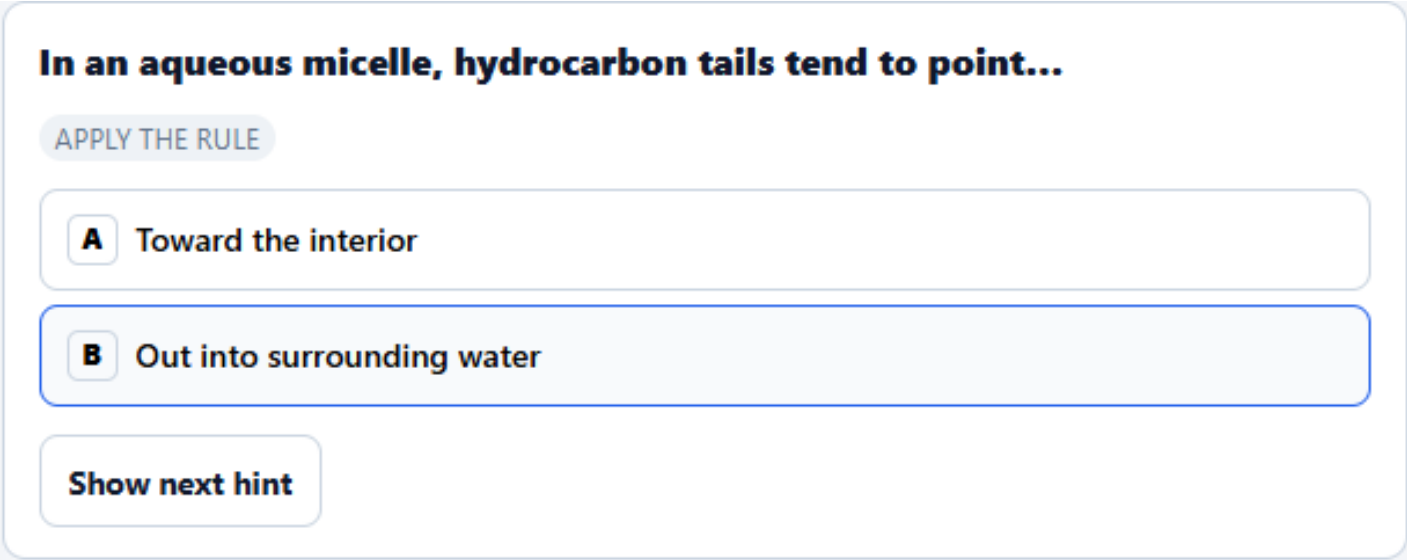
Option 0 (A): Toward the interior

Option 1 (B): Out into surrounding water

Screenshot: soap-head-stage-2-molecule



Screenshot: soap-head-stage-2-question



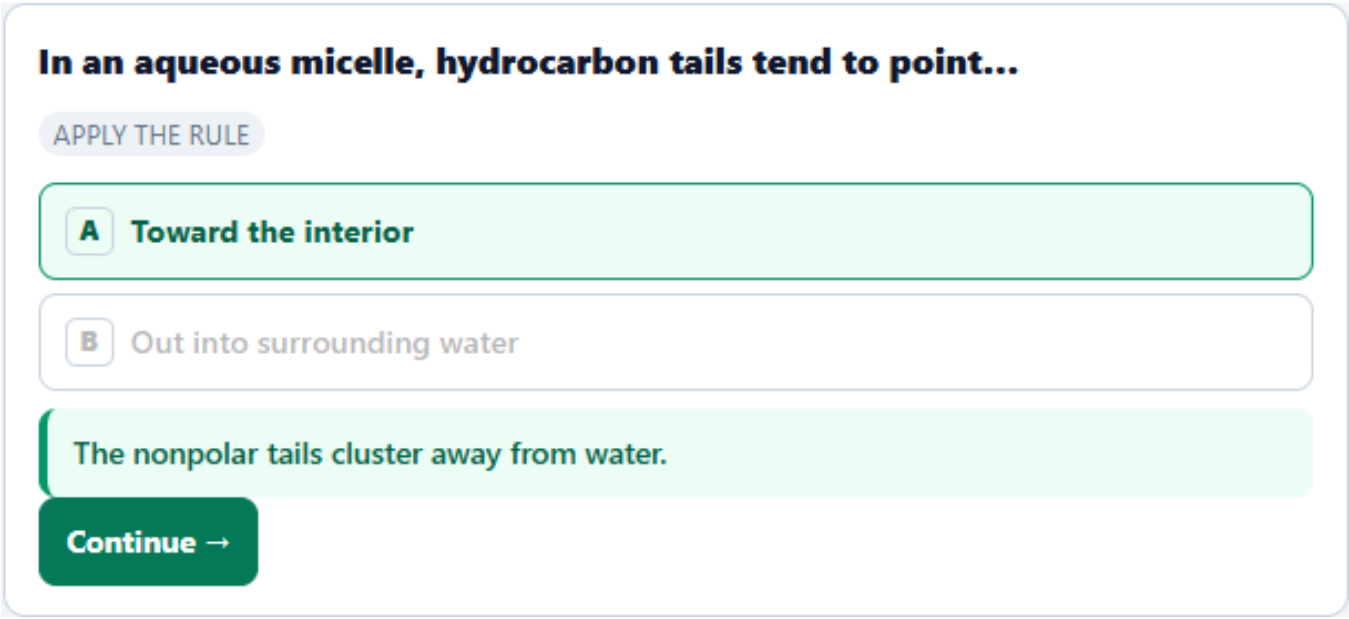
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0, 12, 13]}}

Hint 1: Distinguish polar and nonpolar regions.

Hint 2: Ionic heads remain exposed to water.

Explanation: The nonpolar tails cluster away from water.

Screenshot: soap-head-stage-2-feedback



soap-head:stage-3 | Saponification of a triacylglycerol produces...

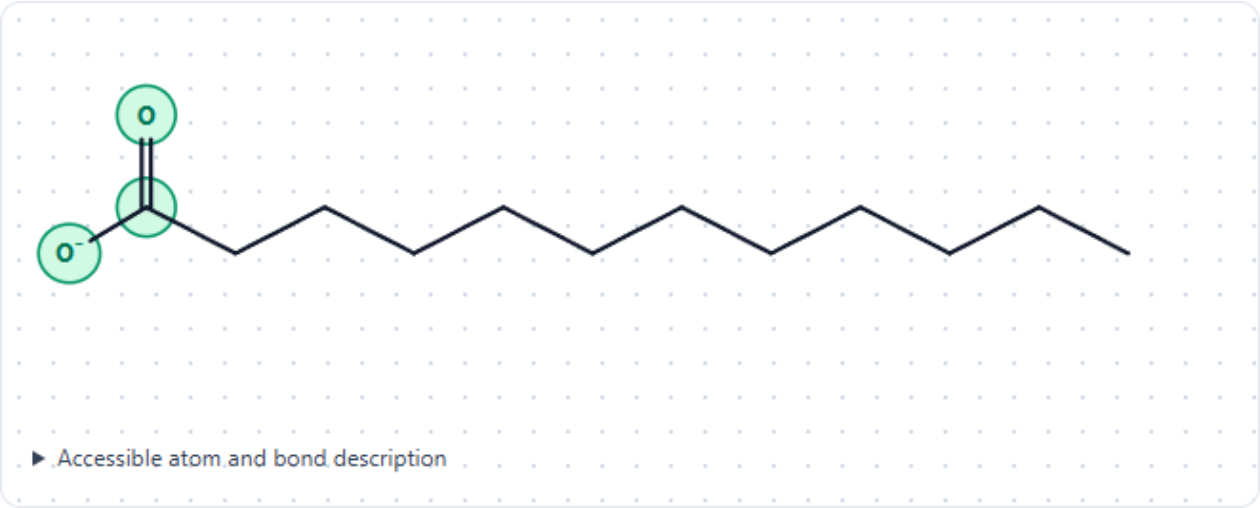
Answer type: choice

Option 0 (A): Glycerol and fatty-acid salts

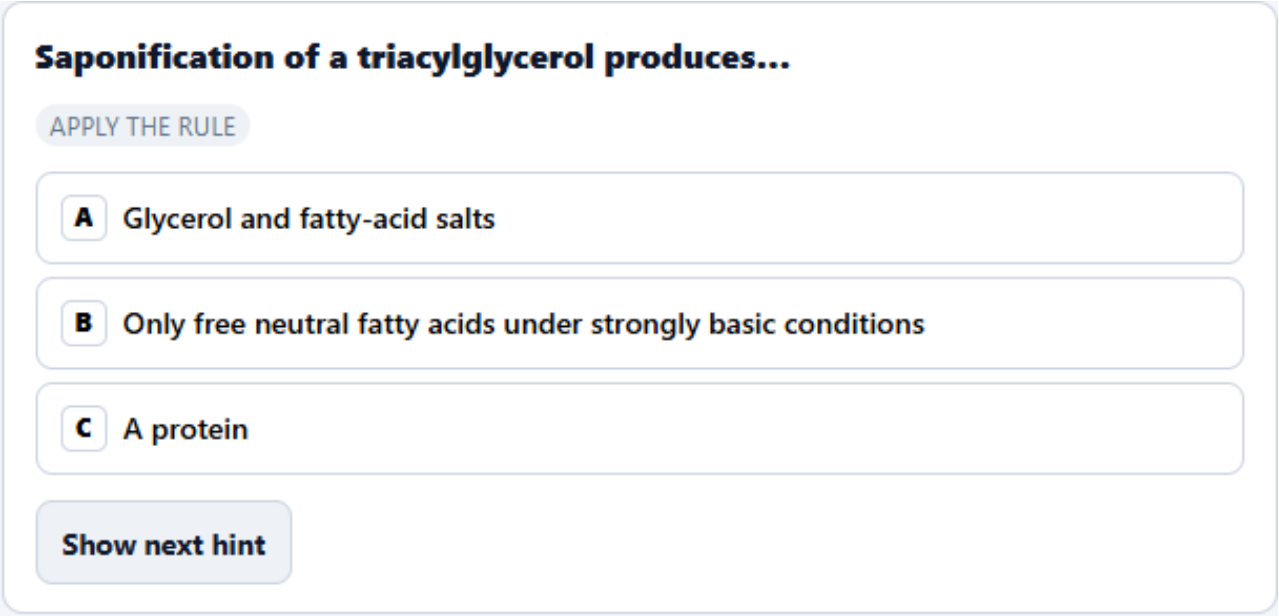
Option 1 (B): Only free neutral fatty acids under strongly basic conditions

Option 2 (C): A protein

Screenshot: soap-head-stage-3-molecule



Screenshot: soap-head-stage-3-question



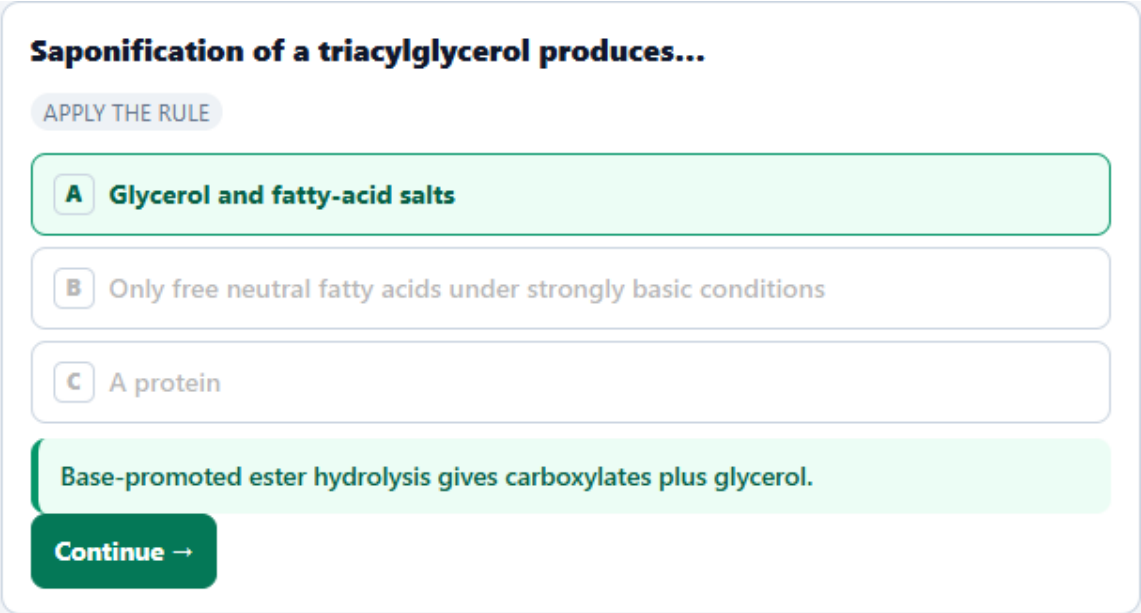
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0, 12, 13]}}

Hint 1: Hydrolyze all three ester links.

Hint 2: Under basic conditions the acids are deprotonated.

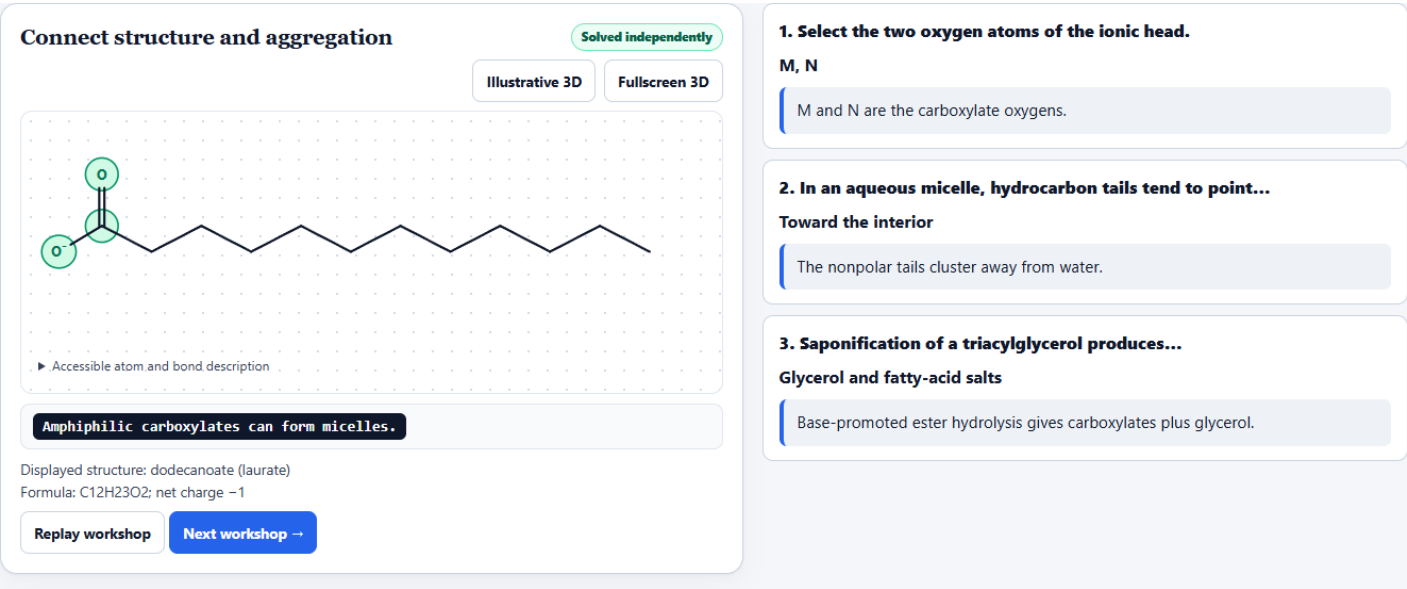
Explanation: Base-promoted ester hydrolysis gives carboxylates plus glycerol.

Screenshot: soap-head-stage-3-feedback



Completion summary: Amphiphilic carboxylates can form micelles.

Screenshot: soap-head-complete



Molecular repository: author claims to verify

aceticacid

Author claims: ethanoic acid | formula C2H4O2 | canonical composition C2H4O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=108.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

acetate

Author claims: acetate (one resonance contributor) | formula C2H3O2 | canonical composition C2H3O2 | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=175.00, y=115.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

ester

Author claims: ethyl propanoate | formula C5H10O2 | canonical composition C5H10O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=156.00, y=52.00; charge=0; explicit H override=none

4 (E): O; x=208.00, y=142.00; charge=0; explicit H override=none

5 (F): C; x=258.00, y=112.00; charge=0; explicit H override=none

6 (G): C; x=308.00, y=142.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [5, 6, 1]]

acetamide

Author claims: ethanamide | formula C2H5NO | canonical composition C2H5NO | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): N; x=175.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

acetoaceticacid

Author claims: 3-oxobutanoic acid (acetoacetic acid) | formula C4H6O3 | canonical composition C4H6O3 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=85.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=115.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=85.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=115.00; charge=0; explicit H override=none

4 (E): O; x=50.00, y=25.00; charge=0; explicit H override=none

5 (F): O; x=0.00, y=115.00; charge=0; explicit H override=none

6 (G): O; x=166.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [0, 4, 2], [0, 5, 1], [2, 6, 2]]

acetone

Author claims: propanone | formula C3H6O | canonical composition C3H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

laurate

Author claims: dodecanoate (laurate) | formula C12H23O2 | canonical composition C12H23O2 | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=85.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=115.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=85.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=115.00; charge=0; explicit H override=none

4 (E): C; x=282.00, y=85.00; charge=0; explicit H override=none

5 (F): C; x=340.00, y=115.00; charge=0; explicit H override=none

6 (G): C; x=398.00, y=85.00; charge=0; explicit H override=none

7 (H): C; x=456.00, y=115.00; charge=0; explicit H override=none

8 (I): C; x=514.00, y=85.00; charge=0; explicit H override=none

9 (J): C; x=572.00, y=115.00; charge=0; explicit H override=none

10 (K): C; x=630.00, y=85.00; charge=0; explicit H override=none

11 (L): C; x=688.00, y=115.00; charge=0; explicit H override=none

12 (M): O; x=50.00, y=25.00; charge=0; explicit H override=none

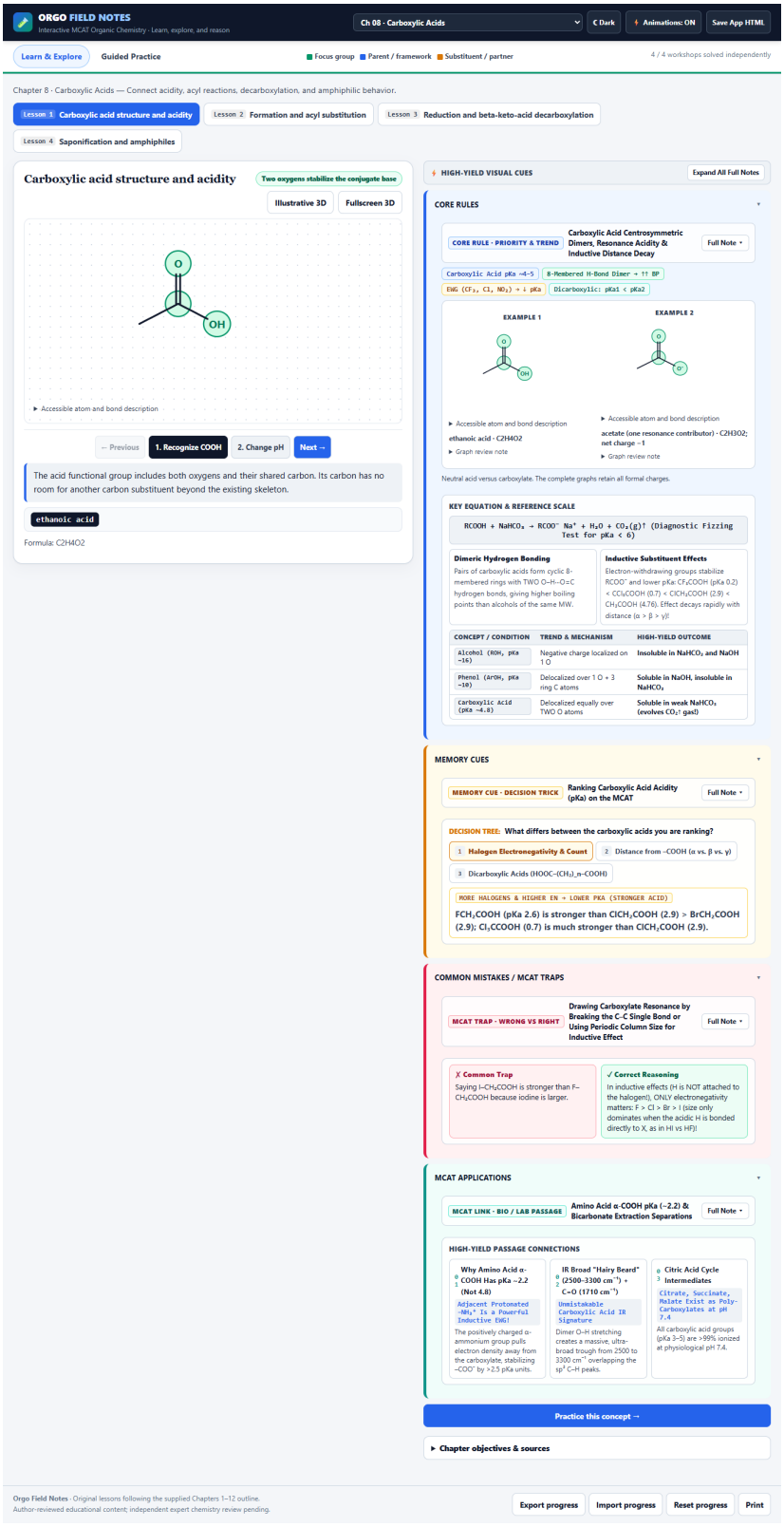
13 (N): O; x=0.00, y=115.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 5, 1], [5, 6, 1], [6, 7, 1], [7, 8, 1], [8, 9, 1], [9, 10, 1], [10, 11, 1], [0, 12, 2], [0, 13, 1]]

Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

Screenshot: layout-desktop-light



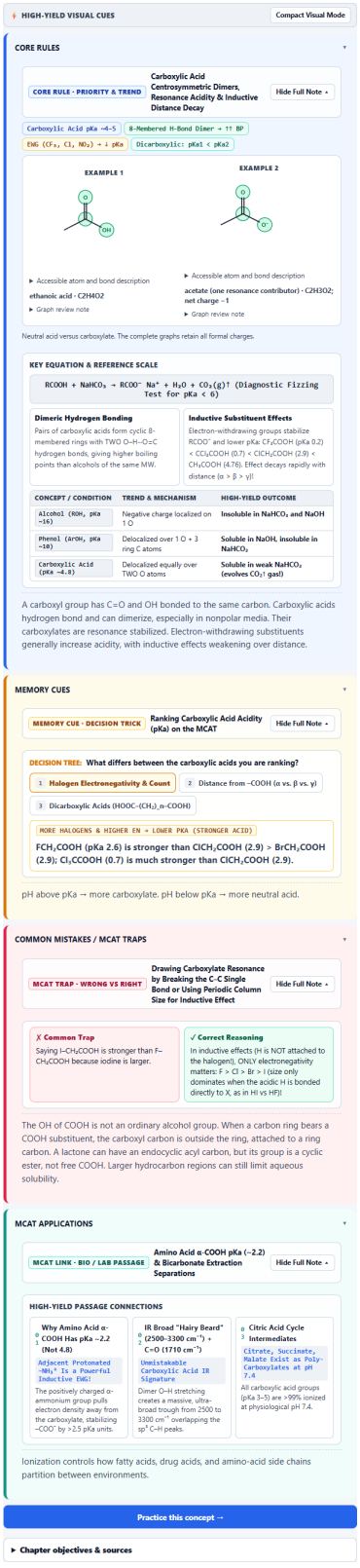
Screenshot: layout-mobile-light



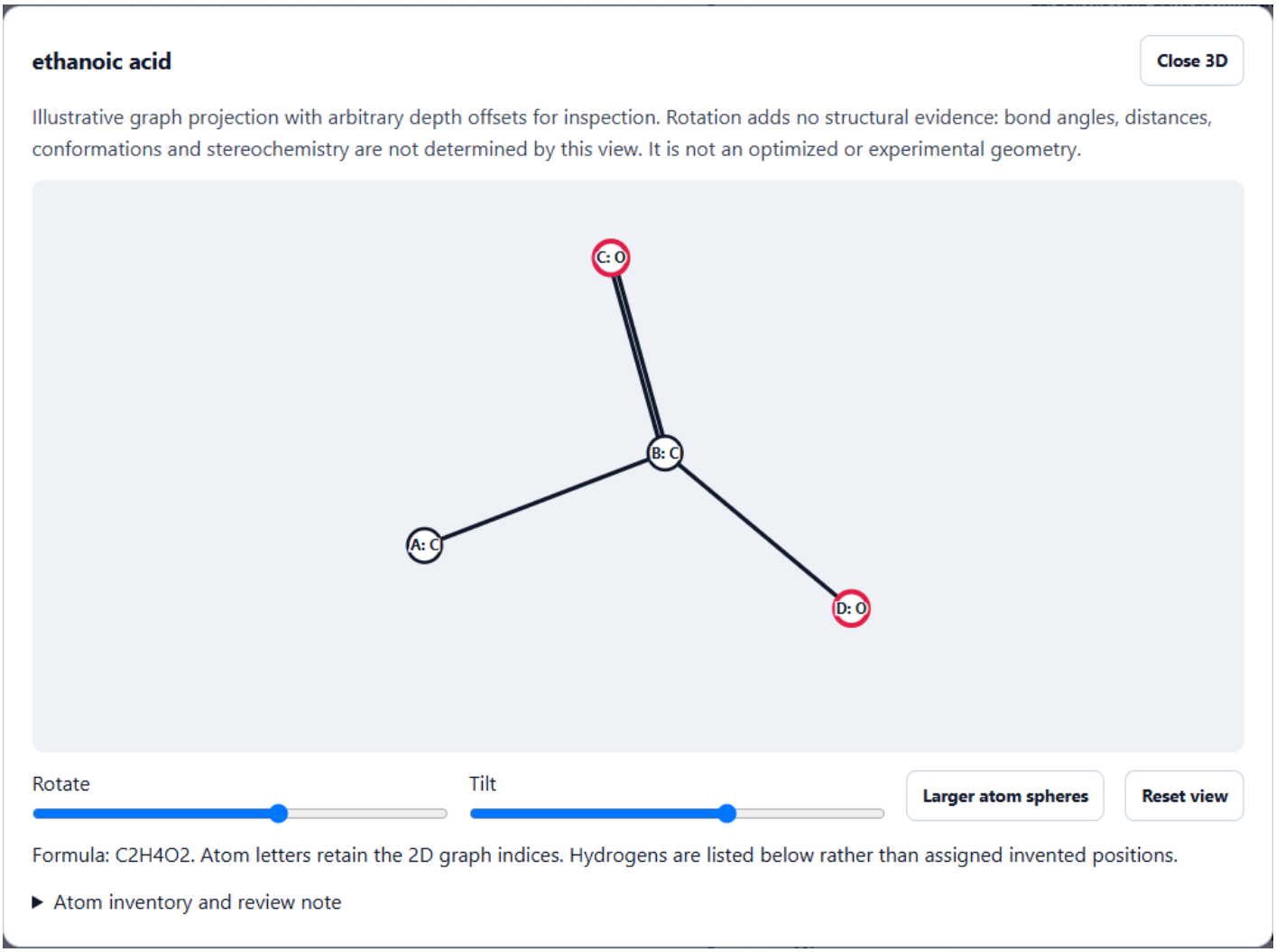
Screenshot: layout-mobile-dark



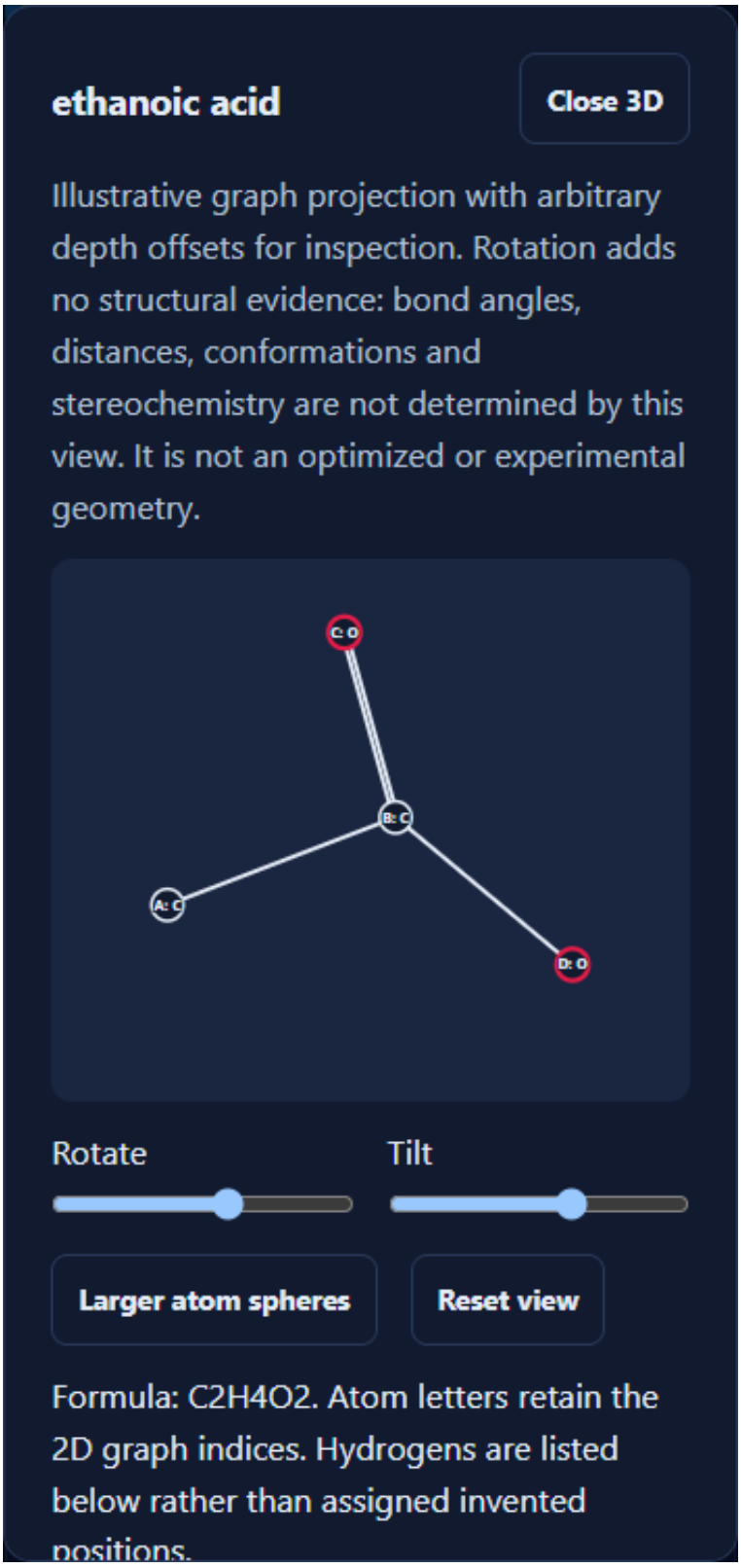
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

60 screenshots and 12 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.